



Systematic mapping of methods for real-time capable multibody simulation of road vehicles using PRISMA

Jan-Lukas Archut¹ · Burkhard Corves¹

Received: 11 December 2024 / Accepted: 30 April 2025

© The Author(s) 2025

Abstract

Road vehicle dynamics studies the performance of road vehicles to improve their driving characteristics using the principles of multibody system dynamics. The growing importance of virtual design methods, especially of real-time applications like driving simulators, raises the demand for more accurate real-time vehicle models. This paper reports a systematic mapping, based on the PRISMA methodology, of scientific publications from 1999 to 2024 presenting real-time vehicle dynamics models and their underlying methods. It includes vehicle models with lumped-mass, lookup-table and multibody-based suspensions, as well as standalone multibody suspension models. Literature was retrieved from Scopus, Web of Science and Google Scholar using keywords targeting real-time multibody simulation of vehicles and suspensions. From a total of 765 records, 109 eligible publications were analyzed in detail. The main findings are: (a) most of the publications consider multibody suspensions and focus on the research of related real-time methods; publications considering models with lumped-mass and lookup-table suspensions mainly focus on their real-time application; (b) the suspension elastokinematics is neglected in most of the multibody suspension models; if included, then mainly through bushing elements, with body flexibility rarely considered; (c) real-time capability can be achieved with various equations of motion formulation and integration methods, depending on the characteristics of the multibody suspension model; (d) diverse model simplifications and approaches in the equations of motion formulation, integration and implementation aim to reduce computation times. This study provides an overview of real-time vehicle dynamics models and methods with detailed explanation of the findings.

Keywords Two-track model · Elastic multibody simulation · Model reduction · Numerical stiffness · Linear-implicit integration · Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)

1 Introduction

Good driving dynamics are an important design objective in vehicle development, because road vehicles should be safe, comfortable and even enjoyable to drive. The suspension sys-

✉ J.-L. Archut
archut@igmr.rwth-aachen.de

¹ Institute of Mechanism Theory, Machine Dynamics and Robotics, RWTH Aachen University, Eilfschornsteinstrasse 18, 52062 Aachen, Germany

tem, alongside the chassis and powertrain, is a key component of road vehicles. It guides the wheels and transmits all wheel forces from the road to the chassis, forming the basis for the vehicle's dynamic behavior, driving safety and comfort [41]. The suspension kinematics determines the position of the wheels relative to the vehicle body and the road, which in turn affects the force transfer between road and vehicle. In addition to their kinematics, suspensions also feature elastokinematic properties [41]. They arise from elastic bushings utilized in the suspension to connect the suspension bodies, as well as from the structural flexibility of the suspension components and the vehicle structure. These properties not only contribute to enhanced ride comfort by means of vibration isolation and shock absorption, but they also affect the handling through compliant wheel position changes related to wheel forces and torques. The suspension design aims either to compensate these wheel position changes, or even to convert them into desired changes in order to improve the vehicle's driving behavior [74].

Virtual design methods gain increasing importance in vehicle development, because they promise a more targeted, cost-effective and safer development process. This in particular includes real-time applications such as software-in-the-loop (SIL) and hardware-in-the-loop (HIL) simulations, as well as virtual test drives on driving simulators. The latter enable the subjective assessment of driving characteristics already in early development stages without the need for physical prototype vehicles. This facilitates driving dynamics development and enables early validation of the driving characteristics. Thus, there is an increasing demand for more detailed real-time capable vehicle models to enhance the applicability and reliability of such real-time applications.

Over the last decades, numerous scientific publications have emerged focusing on real-time vehicle dynamics simulation, particularly emphasizing detailed multibody simulation models. A comprehensive overview of numerical methods in vehicle dynamics, in particular of numerical integration of the equations of motion (EOM) including real-time capable simulation, has been provided by Arnold et al. [4]. They stated that typical vehicle dynamics problems can be transformed into generic problems of numerical mathematics and solved using appropriate numerical methods. In a more recent review by Bruni et al. [13], a broad overview of current challenges in multibody simulation of vehicles, including railway vehicles, road vehicles, motorcycles and bicycles, was given, and their similarities and differences were addressed. Rong et al. [117] provided a detailed review of modeling and solution methods of flexible multibody system dynamics, without emphasizing specific applications. None of these reviews, however, specifically addressed the modeling and real-time simulation of road vehicles with special focus on the suspension as a central subsystem that strongly affects the driving behavior.

The objective of this systematic mapping is therefore to provide an overview of real-time capable vehicle dynamics models and their underlying methods with an emphasis on the suspension system. This includes vehicle models and suspension models that have been published in scientific literature in the last 25 years, i.e., between 1999 and 2024. The following questions are to be answered by this systematic mapping:

- How is the suspension modeled, and to what extent are its kinematic and elastokinematic properties taken into account?
- What methods were used for the formulation and numerical integration of the EOM?
- What model simplifications and what approaches in the formulation and numerical integration of the EOM, including the implementation, were used to reduce computation times and achieve real-time capability?

Overall, this systematic mapping aims to provide an overview of the models and methods found in the scientific literature, along with their limitations. It may also serve as a starting

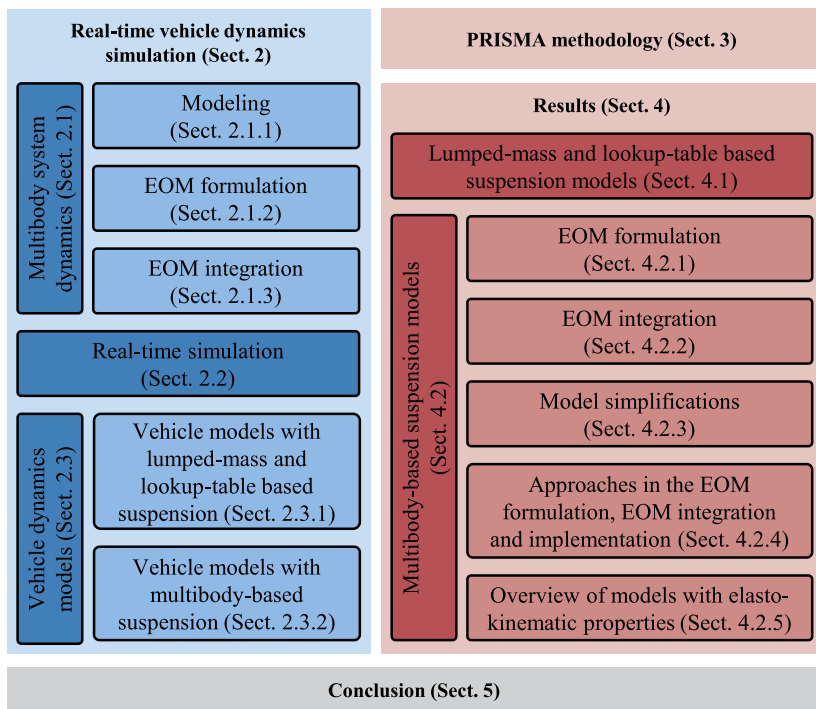


Fig. 1 Conceptual organization of the paper

point for researchers in the field of real-time vehicle dynamics simulation, providing an overview and basic understanding of the key concepts in this research field.

The paper is organized as shown in Fig. 1: the first part provides an introduction into the topic of real-time vehicle dynamics simulation (Sect. 2). For this purpose, first an overview of multibody system dynamics theory relevant for the systematic mapping in this paper is given. Then, the requirements of real-time applications are briefly introduced and discussed. Finally, common vehicle dynamics models and their advantages and disadvantages are presented, motivating the demand for more detailed real-time vehicle dynamics models and illustrating the associated challenges. The second part of the paper forms the core of this work, which is the analysis and systematic mapping of the literature with respect to the aforementioned formulated questions. Section 3 explains the analysis procedure based on the *Preferred Reporting Items for Systematic Reviews and Meta-Analyses* (PRISMA) methodology [85], including the protocol, framework, eligibility criteria, search strategy and processing of identified records. Section 4 presents the results of the systematic mapping. First, an overview of the vehicle dynamics models found in the analyzed publications is given, illustrating their annual distribution and consideration of elastokinematic properties. Next, vehicle models with lumped-mass and with lookup-table based suspension are briefly summarized. Then, real-time multibody simulation methods employed for multibody-based suspensions are analyzed in detail: Sect. 4.2.1 and 4.2.2 address the methods for formulating and numerically integrating the EOM used for real-time simulations. Section 4.2.3 presents the employed model simplification approaches and discusses the trade-off between reduction of computational effort and loss of accuracy. Approaches to reduce computation times in the EOM formulation and integration as well as in the computational implementa-

tion are analyzed in Sect. 4.2.4. Given the increasing demand for incorporating elastokinematic properties into real-time multibody-based suspension models, Sect. 4.2.5 reorganizes the information from the previous sections to highlight the extent to which elastokinematic properties were considered and the associated methods used for real-time simulation. The paper ends with the conclusion given in Sect. 5.

2 Real-time vehicle dynamics simulation

Road vehicle dynamics studies the performance of road vehicles in order to enhance their driving behavior, ride comfort and safety. This includes in particular passenger cars and trucks as the most common types of vehicles on the road. [12, 13]

The foundation for the design and development of vehicle systems and components is built on vehicle dynamics simulation, which enables the prediction and evaluation of a vehicle's dynamic behavior. Driving maneuvers can be simulated as often as required and in a reproducible manner under defined boundary conditions. Parameters and initial conditions can be varied in ways that are not possible in real-life experiments. Furthermore, safety-critical maneuvers can be conducted within a secure virtual environment. These advantages of simulations and virtual prototypes enable ever shorter development cycles for new vehicle series. [125]

Real-time vehicle dynamics simulation connects the virtual world with the real world [3]. Typical applications are HIL/SIL simulations and driving simulators. It has become very important since the first driving simulator was opened in 1985 [13]. Nowadays, it is a standard tool for the design and development of new vehicles [19]. Driving simulators enable virtual test drives in which vehicles in the virtual world are driven by humans in the real world. They make the driving characteristics captured by vehicle dynamics models tangible to the human, allowing for subjective assessments without the need for cost-intensive physical prototype vehicles [35]. Such a closed-loop evaluation of the human-vehicle system is very important, as human behavior and perception depend on the driving situation and are challenging to model [131]. By enhancing the efficiency of vehicle dynamics evaluation, driving simulators have become one of the most effective tools in today's vehicle dynamics development [52, 122].

A key challenge of vehicle dynamics simulation is therefore to develop accurate and valid vehicle dynamics models based on the kinematic and dynamic laws of mechanics and vibration [13].

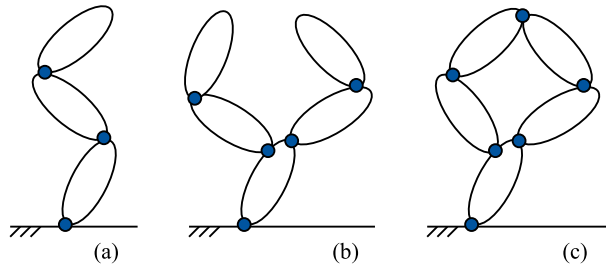
2.1 Multibody system dynamics

The standard framework for ride and comfort simulation of vehicles is multibody system dynamics [1]. It provides the methods for simulation, analysis and optimization of the complex behavior of vehicles and their components [4]. Its main focus of is to evaluate the motion of bodies along with the associated forces and torques [125]. Effectiveness and efficiency of a multibody simulation depend on the modeling of the multibody system, the dynamic formulation used to derive the EOM, and the numerical integration method applied to solve the EOM [28].

2.1.1 Modeling of multibody systems

The modeling of multibody systems involves the determination of the relevant system components and physical effects, the definition of the model elements and their parametrization.

Fig. 2 Topologies of multibody systems (a) Open-loop topology with serial chain; (b) Open-loop topology with tree structure; (c) Closed-loop topology (based on [137])



In general, a multibody system should be modeled complex enough to capture all dynamic effects important for a particular analysis but as simple as possible to reduce unnecessary modeling and analysis efforts and make the system easier to understand. A multibody system consists of a finite number of bodies interconnected by joints and force elements [4]. The bodies are modeled as either rigid or flexible, with the assumption that the mass of the multibody system is concentrated within the bodies. Joints are modeled as massless connections that constrain the free motion of the bodies or impose prescribed motions. Force elements are utilized to model elasticities and damping effects, as well as to apply external forces or actuator forces. The interconnection of the bodies through joints determines the system's topology as open-loop or closed-loop, see Fig. 2 [125]. Open-loop topologies consist of serial chains or tree structures in which each body is connected to only one preceding body. In a closed-loop topology, additional joints are introduced to create at least one closed-loop within the structure.

2.1.2 Formulation of the EOM

The mathematical description of the kinematics and kinetics of a multibody system results in the EOM that describe its dynamic behavior. The EOM take different forms depending on the choice of coordinates, whether independent or dependent. If a system is described through a minimal set of appropriate coordinates $\mathbf{q}_i(t) \in \mathbb{R}^{n_i}$, whose number n_i is equal to the number of degrees of freedom (DOF), then all coordinates are independent of each other. The EOM are then formulated as second-order ordinary differential equations (ODE)

$$\mathbf{M}(\mathbf{q}_i)\ddot{\mathbf{q}}_i(t) = \mathbf{f}(\mathbf{q}_i, \dot{\mathbf{q}}_i, t) \quad (1)$$

with initial values $\mathbf{q}_{i,0} = \mathbf{q}_i(t_0)$ and $\dot{\mathbf{q}}_{i,0} = \dot{\mathbf{q}}_i(t_0)$, which can also be written in state-space form [4]. Here, $\mathbf{M}$ is the positive definite mass matrix, and $\mathbf{f}$ denotes the vector of generalized forces containing applied and gyroscopic forces. Formulations in dependent coordinates $\mathbf{q}(t) \in \mathbb{R}^n$ employ a set of coordinates that contains more than the minimum number of coordinates. These coordinates are redundant and must be related to each other through n_g constraint equations $\mathbf{g}(\mathbf{q}, t) = \mathbf{0}$. They are added to the differential equations, leading to the descriptor form of the EOM as differential-algebraic equations (DAE),

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}}(t) + \mathbf{G}^T(\mathbf{q}, t)\lambda = \mathbf{f}(\mathbf{q}, \dot{\mathbf{q}}, t) \quad (2a)$$

$$\mathbf{g}(\mathbf{q}, t) = \mathbf{0} \quad (2b)$$

with initial values $\mathbf{q}_0 = \mathbf{q}(t_0)$, $\dot{\mathbf{q}}_0 = \dot{\mathbf{q}}(t_0)$ and $\lambda_0 = \lambda(t_0)$ consistent with the constraints [4]. Satisfaction of the constraint conditions is ensured through the constraint forces $\mathbf{G}^T(\mathbf{q}, t)\lambda$

that are determined with the constraint Jacobian matrix or simply constraint Jacobian $\mathbf{G}$ and Lagrange multipliers $\lambda \in \mathbb{R}^{n_g}$.

There are two major families of methods for formulating the EOM of a given multibody system: Global formulations treat all multibody systems in the same way without taking into account the system's specific topology, whether it is open- or closed-loop, or loosely or severely constrained. They use a maximum set of coordinates, usually absolute coordinates, which directly describe the translations and rotations of each individual body. The dynamic terms are calculated independently for each body and then assembled to form the corresponding equations of the whole system. Global formulations are very convenient to use and implement because of their simplicity and generality. They usually lead to differential-algebraic EOM, which are characterized by a high number of dependent coordinates and constraint equations. Therefore, global formulations may experience inefficiencies in performance. [22, 26, 44, 83, 99, 105]

Topological formulations take the specific topology of the multibody system into account and employ relative joint coordinates, which describe the kinematics of a body relative to the previous body in a kinematic chain. For systems with open-loop topology, these coordinates constitute the minimum set of coordinates, and the EOM are formulated as ODE. Closed-loop topologies are usually first transformed into an open-loop topology by means of joint-cutting or similar techniques, and the EOM are subsequently augmented by constraint equations to satisfy the loop-closure conditions. This leads to differential-algebraic EOM with, in comparison to global formulations, fewer dependent coordinates and loop-closure constraint equations. Consequently, topological formulations result in small-size ODE or DAE by taking advantage of the system's topology. However, they tend to be complex and difficult to generalize. [26, 44, 83]

Differential-algebraic EOM can be transformed into ordinary differential EOM through coordinate reduction to independent coordinates. The coordinate partitioning method exploits the null space of the constraint Jacobian and partitions the matrix into dependent columns related to a subset of dependent coordinates, and independent columns related to a subset of independent coordinates [46]. From this, a transformation matrix is derived that reduces the original dependent coordinates onto the constraint manifold and eliminates the unknown constraint forces and Lagrange multipliers from the EOM [133]. The partition into dependent and independent coordinates can be performed numerically through Gaussian elimination and column pivoting [46]. This partition may only be locally valid, requiring a computationally expensive repartitioning if the constraint manifold changes. Alternatively, independent coordinates can be selected manually based on the system's topology [126].

2.1.3 Numerical integration of the EOM

The EOM, either formulated as ODE or DAE, are usually nonlinear and form an initial value problem, which is solved in the time domain with appropriate numerical integration methods. A comprehensive overview of numerical solution methods for ordinary differential and differential-algebraic EOM in vehicle system dynamics was provided by Arnold et al. [4]. The appropriate method must be selected for a specific problem with respect to certain requirements. It must provide a sufficiently accurate solution of the EOM, along with sufficiently low computation times. In addition, the solution method must solve the EOM in a numerically stable manner. This is particularly challenging for EOM with high numerical stiffness [4]. There exists a wide variety of integration methods with different properties and advantages as well as disadvantages with respect to these requirements.

Explicit methods compute the numerical solution of the EOM in the next unknown time step through explicit calculations [4]. They require the evaluation of the EOM only for

known states and have constant computational costs per integration step. Unfortunately, explicit methods suffer from significant drawbacks due to their limited stability properties. In particular, stable integration of numerically stiff EOM can only be achieved by reducing the integration step size to unreasonably small values [4].

Implicit methods compute the numerical solution of the EOM in the next time step with an implicit equation that involves the evaluation of the EOM in that unknown time step [4]. They have better stability properties and robustness [4], making them well suited for integrating numerically stiff EOM in a stable manner. However, the implicit computation involves nonlinear equation systems that are typically solved with iterative procedures such as the Newton method. This implies higher computational costs per integration step compared to explicit methods. Furthermore, the number of iterations required to achieve a certain error threshold is not a priori known, meaning that the computational costs per integration step may vary during simulation depending on the state and excitation of the multibody system.

Integration of differential-algebraic EOM with ODE-solvers is based on a prior transformation of the EOM [4]. The differential-algebraic EOM (2a)–(2b) are index-3 DAE with constraint equations on the position level. This constraint condition implies additional restrictions on the velocity and acceleration level [4]. Through differentiation of equation (2b) with respect to time t , one obtains the constraint equations on the velocity level

$$\frac{d}{dt} \mathbf{g}(\mathbf{q}(t), t) = \frac{\partial \mathbf{g}}{\partial \mathbf{q}} \frac{d\mathbf{q}}{dt} + \frac{\partial \mathbf{g}}{\partial t} = \mathbf{G}\dot{\mathbf{q}}(t) + \tilde{\mathbf{g}} = \mathbf{0} \quad (3)$$

and on the acceleration level

$$\frac{d^2}{dt^2} \mathbf{g}(\mathbf{q}(t), t) = \mathbf{G}\ddot{\mathbf{q}}(t) + \dot{\mathbf{G}}\dot{\mathbf{q}}(t) + \dot{\tilde{\mathbf{g}}} = \mathbf{G}\ddot{\mathbf{q}}(t) + \tilde{\tilde{\mathbf{g}}} = \mathbf{0}, \quad (4)$$

all of which must be satisfied by the solution of the DAE [4]. The DAE can be transformed analytically before time integration by means of index reduction, substituting the constraints on the position level by one of its time derivatives [4]. They are then called to be in index-2 or index-1 form, depending on whether the constraints on the velocity level or on the acceleration level are considered. Long-term simulations however may suffer from a drift-off of the solution from the constraint manifold, because only the derivatives of the constraint equations are explicitly considered in the solution. Accumulation of the constraint violations can be prevented with stabilization methods like Baumgarte stabilization, which utilizes a linear combination of the constraint equations and their derivatives [4, 10]. They are weighted through so-called Baumgarte coefficients, which have to be chosen specifically for the given problem. Projection techniques are another method to prevent the drift-off effect and project the numerical solution onto the constraint manifold during the integration process [4]. Alternatively to index reduction, the DAE can be transformed into an augmented Lagrangian form [11]. The constraint conditions are represented in the EOM by a penalty term that is added to equation (2a), rather than using separate constraint equations. The unknown Lagrange multipliers are iteratively determined in each integration step, and numerical drift-off is prevented through projection on the constraint manifold.

2.2 Real-time simulation

According to the standard ISO/IEC 2382:2015, the term real-time refers to “the processing of data by a computer in connection with another process outside the computer according to time requirements imposed by the outside process” [51]. Real-time can be further classified into hard real-time and soft real-time [64], see Fig. 3: Hard real-time requires that the

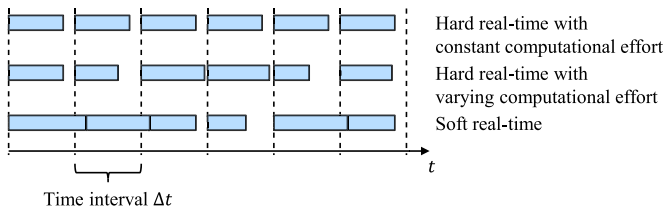


Fig. 3 Hard and soft real-time

computation results must be generated within the defined time interval Δt at any time. The computation time per interval may vary, but exceeding the time interval is never permitted. This is the case in driving simulators and HIL applications. In soft real-time, the defined interval may be exceeded in rare events, as long as adherence to the time interval is statistically guaranteed. Examples are web applications or live video transmission [65, 123].

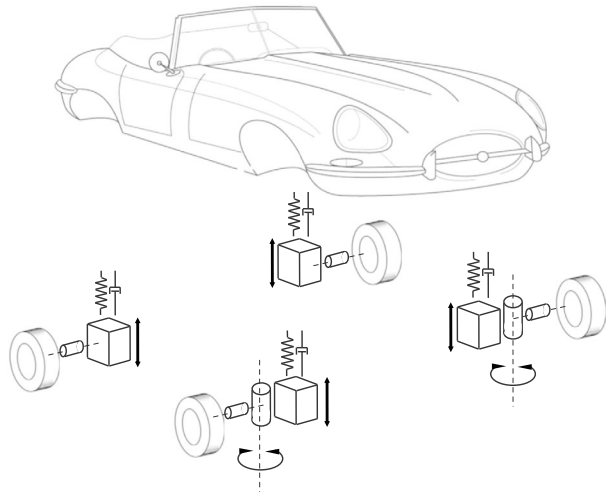
These definitions show that real-time capability is always related to the application. In real-time vehicle dynamics simulations, the EOM of the multibody model must be evaluated and numerically solved in each time step within the fixed time interval that is dictated by the real-time application [4]. This time interval is typically 1 ms as the current standard for automotive HIL applications [15, 19, 108] and a typical value for driving simulators [65]. It is important that the turnaround time in each time step is a priori known to be less [45]. Therefore, explicit integration methods are attractive for real-time applications, as they require constant computational costs in each integration step. In contrast, implicit integration methods are usually less well suited due to their higher and varying computational costs per integration step. Due to the step size restrictions of hard real-time applications, integration step sizes cannot be increased arbitrarily to compensate for these higher computational costs. Convergence of the iterative solution within the time interval may fail in critical situations [140]. A demanding vehicle dynamics simulation may therefore fail in these applications and may only be able to run in real-time for larger time intervals. However, more performant computer hardware could enable real-time simulation even for smaller time intervals.

2.3 Vehicle dynamics models

Vehicle dynamics models of varying complexity and sophistication are used for vehicle dynamics simulation, depending on the task and level of detail required. Simulation of general driving maneuvers, including longitudinal, lateral and vertical dynamics, requires two-track models that cover all driving dynamics domains. Such a two-track model has at least 14 DOF, consisting of all six DOF for the vehicle body, four DOF for the wheel travel and four DOF for the wheel rotation. Steering motion and drive/brake torques are considered as inputs. The suspension can be modeled at different levels of detail. Frequently used are two-track models with lumped-mass (LM) suspension, lookup-table based (LT) suspension, and multibody-based (MBS) suspension. Two-track models are completed with models of the tire and the road and may be complemented by further subsystems such as steering system, engine, drive train, driver and loads to model the full vehicle [12, 112]. In the following, they will also be referred to as vehicle models.

A wide range of simulation tools and software packages for vehicle dynamics simulation is available today, including complete simulation environments and tool chains. They can be divided into multi-purpose and special-purpose packages [1, 13]. The first group is not specifically designed for vehicle dynamics simulation but is suitable for all multibody systems [1]. Complex, non-linear and three-dimensional vehicle models, incorporating flexible

Fig. 4 Vehicle model with LM-suspension ([12], modified)



bodies, can be created for handling and comfort analysis [13]. Typical examples include Adams, SIMPACK, RecurDyn and Dymola [13]. Special-purpose packages focus on automotive applications and provide model templates with ready-to-use vehicle models and tailored evaluation methods [13]. These can be non-linear, three-dimensional vehicle models for both offline and real-time simulations [13]. Real-time performance is typically achieved through lookup-table based modeling of the suspension [13]. Typical examples are CarSim, CarMaker and veDyna [13]. Many multi-purpose software packages also offer automotive-related extensions, such as Adams/Car, SIMPACK Automotive and Virtual.Lab VehicleMotion [1]. In general, commercial packages provide low compatibility and interfaces with third-party software tools [13]. Furthermore, the underlying methods and implementations are usually only accessible to a limited extent and not fully published [65]. For these reasons, also “home-made” multibody models are developed and used in industrial applications and for research purposes [13].

2.3.1 Vehicle models with lumped-mass and lookup-table based suspension

LM-suspension models simplify the suspension system to a single mass that can only perform translational motions in vertical direction relative to the vehicle body [12, 125]. A vehicle model with LM-suspension consists of the vehicle body, which accounts for the complete vehicle’s sprung mass, and four lumped masses representing the unsprung masses of the suspension components, see Fig. 4 [12]. The lumped masses are guided relative to the vehicle body by means of prismatic joints, disregarding the actual kinematics and elastokinematics of the suspension. However, the prismatic joints may be tilted to account for the anti-dive and anti-squat geometry and roll center height of the suspension [115]. The main spring and damper of the suspension are placed in parallel to the prismatic joints and have linear- or nonlinear properties [25]. The wheel bodies are connected to the lumped suspension masses through revolute joints that allow spinning of the wheels. The steering is enabled by additional revolute joints with vertical rotation axes and usually assumed as prescribed input.

Vehicle models with LT-suspension extend the LM-suspension approach with a simplified representation of the suspension kinematics, and optionally elastokinematics, by means

of lookup-tables [1]. These lookup-tables contain information about wheel position and orientation changes or other kinematic parameters as a function of the wheel travel and steering input, and in the case of elastokinematic properties, also as a function of the wheel forces and torques [18]. They are obtained from kinematics and compliance (K&C) tests. Typically, a model simplification process based on detailed multibody vehicle models is performed [18]. Several, usually quasistatic K&C tests are conducted, in which the vehicle body is fixed and inputs such as parallel and opposite wheel travel, steering, and different wheel forces and torques are applied [8, 18, 145]. Wheel alignment and other suspension parameters are recorded and stored in the lookup-tables. Alternatively, polynomial functions describing these parameters are obtained using curve fitting methods [146]. This model simplification process is usually automated, and automatic conversion tools are provided by dedicated commercial software [8, 18, 145]. K&C tests can also be conducted experimentally with real vehicles on suspension parameter measurement machines [7, 36].

Vehicle models with LM- and LT-suspension provide several benefits for real-time applications. They are characterized by a low complexity, featuring a simple open-loop topology with tree structure, low number of DOF and, at least for LM-suspension models, only a few parameters required [18]. They are inherently real-time capable [24, 62], and are implemented in many commercial special-purpose software packages. LT-suspension models allow to analyze any type of suspension regardless of its specific design, requiring mainly a modified parametrization of the lookup-tables [145]. Elastokinematic properties can be taken into account up to a certain degree of accuracy, while preserving real-time capability [18]. Vehicle models with LT-suspension demonstrate a good accuracy in the frequency range up to approximately 5 Hz, which covers the typical roll, pitch and heave motion of the vehicle [108]. Comparisons with detailed vehicle models for different handling maneuvers can be found in [18, 146].

However, LM- and LT-suspension models also exhibit serious drawbacks. As the physical suspension components are not explicitly modeled, any change of their parametrization requires a regeneration of the lookup-tables, which can become a cumbersome task [8, 19, 77, 108, 145]. Furthermore, internal quantities of the suspension system, e.g., the forces in the joints or bushings, are not accessible, limiting the informative value of the models [23, 138, 139]. LT-suspension models cover the kinematics and elastokinematics of the suspension, but only up to a certain degree of accuracy. The lookup-tables are generated based on a limited number of K&C tests and describe the wheel motion as a linear superposition of the effects due to wheel travel, steering and wheel forces and torques [8]. Thus, interaction effects, e.g., different suspension compliance at different wheel travels, are simplified. Furthermore, the internal dynamics of the suspension is neglected, as the K&C tests are typically performed quasistatically. These drawbacks limit the ability of the models to perform realistic vehicle dynamics simulations, because correlation with more detailed vehicle models deteriorates at higher frequencies [24, 42]. For highly dynamic driving maneuvers or ride comfort scenarios with typical frequencies up to 30 Hz, extensions of the models or even vehicle models with MBS-suspension are required [8, 42, 77].

2.3.2 Vehicle models with multibody-based suspension

An MBS-suspension model represents the actual suspension assembly as a proper multibody system with bodies, joints and force elements, see Fig. 5, in order to reproduce the behavior of the real system as closely as possible [12, 112, 125]. Their kinematic and dynamic properties including positions, masses, moments of inertia, stiffness and damping characteristics are explicitly modeled and described through parameters with actual physical meaning

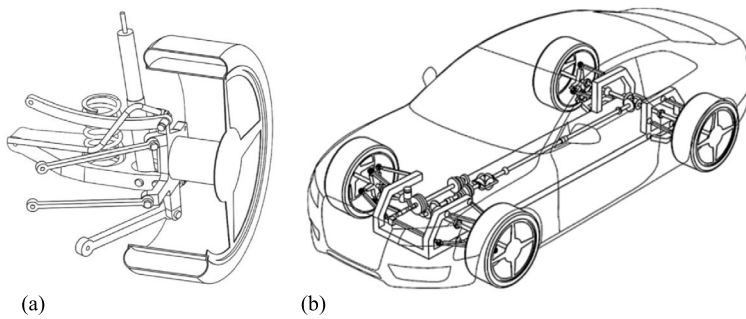


Fig. 5 Multibody modeling of suspensions (a) MBS-suspension model of a multilink suspension; (b) Vehicle model with MBS-suspension [125]

[95]. In a purely kinematic MBS-suspension model, the bodies are assumed to be rigid and connected with each other through ideal joints [112]. Main spring and damper of the suspension are modeled as linear or nonlinear force elements. In a more detailed elastokinematic MBS-suspension model, elastokinematic properties are introduced by means of bushings and flexible bodies [12, 112]. Bushings are typically a combination of stiff nonlinear translational and torsional spring-damper elements [12, 59]. Depending on the level of fidelity required, more complex bushing models with detailed frequency-dependent properties or hydro-mount models of advanced rubber-to-metal liquid elements can be used, e.g., for extensive ride comfort analyses [42, 66]. Bushings replace ideal kinematic joints between the suspension bodies, so that elastokinematic MBS-suspension models often have more than 50 DOF [108].

Vehicle models with MBS-suspension offer numerous advantages. They take the effects of the suspension topology directly into account, including the motion of all suspension bodies and their interaction, as well as the nonlinearities of the suspension system [95, 129]. They provide more accurate and reliable simulation results compared to vehicle models with LM- or LT-suspension [45, 58, 99]. Model validity is extended to higher frequencies, including the dynamic effects related to the elastokinematics [108, 146]. Real-time capable vehicle models with detailed MBS-suspension increase the capabilities of virtual vehicle dynamics analysis, e.g., they extend the informative value of driving simulators from handling assessments to ride comfort evaluation [77]. They also bridge the gap between detailed vehicle models and real-time applications, eliminating the need for parameter conversion and lookup-table preparation in contrast to LM- and LT-suspensions [67]. Suspension design parameters can be modified directly, facilitating extensive parameter studies [45, 67]. In addition, all internal variables such as loads in the suspension are accessible for design optimization or further analyses [45]. To exploit all these benefits and avoid the drawbacks associated with LM- and LT-suspensions, real-time capable vehicle models with detailed MBS-suspensions are desired [19].

Achieving real-time capability with MBS-suspension models is challenging for several reasons. Firstly, they are characterized by a high number of interconnected bodies, along with a large number of coordinates and equations [58, 108]. This complexity is further increased by the inclusion of elastokinematic properties, complex force elements or control systems. Secondly, suspension systems are often formed by closed kinematic chains. Accounting for constraints leads to complicated differential-algebraic EOM, which make it difficult to achieve real-time capability for small integration step sizes [58]. Thirdly, detailed vehicle models include components with high stiffness and/or damping properties,

e.g., bushings or flexible bodies that account for the suspension's elastokinematics. These components induce high-frequency or strongly damped motions, as opposed to the slow dynamics of, e.g., the wheel travel motion. As a result, the model exhibits dynamics on different time scales. From a numerical point of view, this is associated with high numerical stiffness of the EOM and makes stable and real-time capable integration challenging [59, 129, 132]. The demand for detailed vehicle dynamics models for real-time applications poses therefore significant challenges, because more complex models lead to increased computation times.

3 PRISMA methodology

This systematic mapping was performed in accordance with the *Preferred Reporting Items for Systematic Reviews and Meta-Analyses* (PRISMA) [85]. The guidelines of this methodology consist of a 27-item checklist and a flow diagram, and provide explicit means for identification, selection, and critical evaluation of studies relevant to the defined research objective. They help to improve reporting of systematic reviews and meta-analyses, and can also be applied to systematic mappings [76].

Publications included in our systematic mapping have to meet various pre-defined eligibility criteria. Only articles, full conference papers, reviews and books or book chapters in English language published between 1999 and 2024 were considered. The publications must contain a two-track model of a car, truck or bus that covers all driving dynamics domains and is suitable for general driving maneuvers. The model must feature a suspension that is modeled as LM-, LT-, or MBS-suspension, with the type of suspension modeling clearly indicated. In addition, publications that contain stand-alone MBS-suspension models were also considered. The models must be used for real-time capable forward dynamics simulation with numerical time integration. Real-time capability must be proven by computation time measurements, explicitly stated, or clearly evident from the application of the model in a real-time environment. Only two-track models with LM- or LT-suspension were assumed to be real-time capable without such proof, as they are inherently real-time capable because of their simple topology [24, 62].

Scopus database, Web of Science and Google Scholar were utilized to search for publications. The search was performed with the keywords *real-time*, *multibody*, *vehicle*, *suspension*, *driv**, *handl** and *automo**, which were combined to the following queries:

- In Scopus:

TITLE-ABS-KEY((realtime OR "real-time" OR "real time") AND (multibody OR "multi body" OR "multi-body") AND (vehicle OR suspension OR driv OR handl* OR automo*))*

- In Web of Science:

(ALL=(realtime) OR ALL=("real time") OR ALL=("real-time")) AND (ALL=(multibody) OR ALL=("multi body") OR ALL=("multi-body")) AND (ALL=(vehicle) OR ALL=(suspension) OR ALL=(driv) OR ALL=(handl*) OR ALL=(automo*))*

In Google Scholar, because of its limited search capabilities, publications were searched for by title only, using all feasible combinations of the keywords mentioned above. Records found were added to the search results if they were not indexed in Scopus or Web of Science. The previously mentioned criteria on publication type, year of publication and language were also considered in the queries. The search was performed on 22 May 2024.

The search yielded 428 records found in Scopus, 326 records found in Web of Science, and 11 additional records found in Google Scholar, which made a total of 765. The records

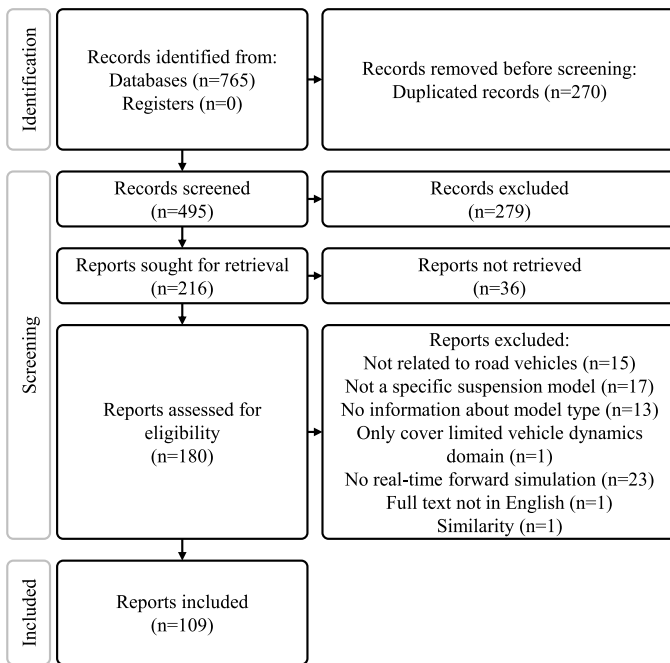


Fig. 6 Flow chart of study selection process (adapted from the template in [85])

were exported and merged into an Excel sheet, including authors, title, abstract, keywords, year of publication and other metadata. Figure 6 depicts the applied selection process based on the PRISMA statement, which is divided into the phases of *identification*, *screening* and *included*. In the *identification* phase, 270 records were removed as duplicates having the same Digital Object Identifier (DOI) and/or title. Then, in the *screening* phase, titles and abstracts of the remaining 495 records were perused. 279 records were excluded in this stage because they clearly did not meet the eligibility criteria. This included, among others, publications related to vehicles such as motorcycles, bicycles, railway vehicles, aircrafts and aerospace vehicles, watercrafts and underwater vehicles, or related to mechanisms and robots, drive trains, hydraulics, tire models and terrain/soil models. Of the remaining 216 reports sought for retrieval, 36 could not be retrieved. Specifically, 22 could not be retrieved because of restricted online access, seven could not be found when searching for by title in Google, and for one report, no pdf document was available. Furthermore, only short rather than full papers were available for six conference contributions, leading to their exclusion. The full texts of the remaining reports were downloaded as pdf documents and saved in the commercial reference management software Citavi. These 180 reports were assessed for eligibility by analyzing their full texts. At this stage, 15 reports were excluded because their content was not related to road vehicles. 17 reports did not contain a specific model with a suspension, and in the case of 13 reports, there was no information about the type of suspension model. One report covered only the vertical driving dynamics domain, and in 23 reports with vehicle models including suspension, there was no explicitly stated real-time capable forward simulation. Additionally, one report was excluded because the full text was in another language despite Scopus indicating it in English. Lastly, one publication was excluded because it had strong similarities to another, previously published publication

by other authors. Consequently, all these reports were systematically excluded from the analysis. A final set of 109 *included* reports was considered for the systematic mapping.

4 Results

This section presents the information found in the included reports that pertains to the objective of this systematic mapping. Figure 7 shows the chronological distribution of publications related to LM-/LT-suspensions and MBS-suspensions. Most of the included reports consider vehicle models with MBS-suspension or stand-alone MBS-suspension models, with a total of 84 publications. A first peak in the number of publications can be observed in the mid-2000s, when a comparatively high number of MBS-suspension models without elastokinematic properties was published. Another concentration of MBS-suspension models without elastokinematic properties can also be noted around 2020. Consideration of elastokinematic properties started to increase in the late 2000s and continues to this day, with a peak of five publications in 2015. In comparison, publications containing LM- or LT-suspension models are less frequent, with a total of 30 publications. These publications were published mainly after 2010, with increased consideration of elastokinematic properties in the last years.

A closer look at the extent to which elastokinematic properties are taken into account in the published suspension models is given in Fig. 8. About 43% of the presented models in the category of LM- and LT-suspension models feature elastokinematic properties. The remaining part is divided between LT-suspension models that only represent the suspension kinematics, and LM-suspension models. Regarding MBS-suspension models, almost two thirds of the publications do not consider elastokinematic properties. Those that do account for elastokinematic properties use various combinations of bushings, flexible suspension bodies or flexible vehicle body. Notably, nearly half of these models include only bushings. The combined influence of bushings and flexible suspension bodies is considered in only two publications.

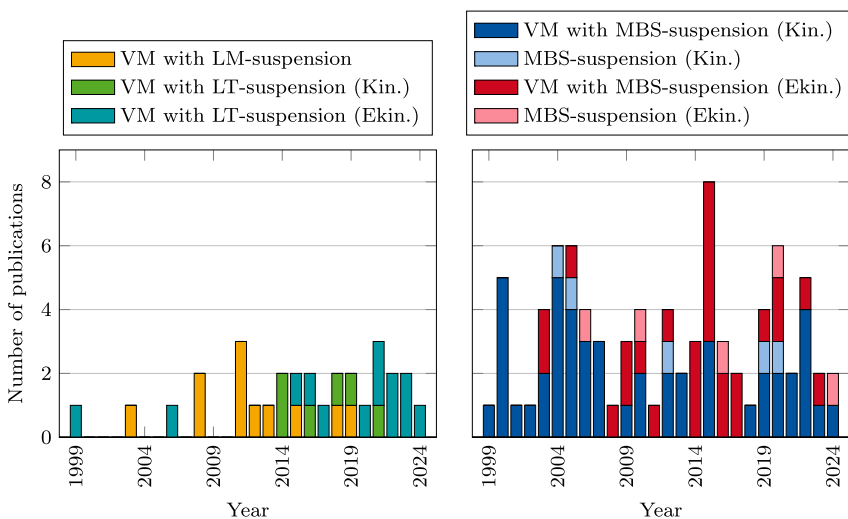


Fig. 7 Type of vehicle and suspension models across time (VM: vehicle model; Kin.: only consideration of the suspension's kinematics; Ekin.: also consideration of the suspension's elastokinematics)

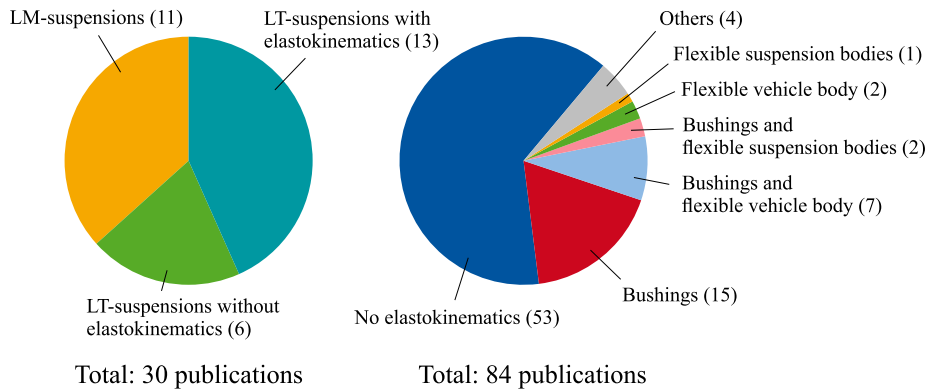


Fig. 8 Detailed view on the consideration of elastokinematic properties (left: vehicle models with LM- and LT-suspension; right: vehicle models with MBS-suspension and MBS-suspension models)

Summarizing these statistics, the majority of the suspension models in the analyzed publications are MBS-suspension models and do not feature any elastokinematic properties. Increased research on the consideration of elastokinematic properties can be observed in the last years.

4.1 Lumped-mass and lookup-table suspension models

The analyzed publications related to LM-suspension models, [14, 25, 39, 73, 75, 101, 103, 110, 115, 136, 140], and to LT-suspension models, [5, 7–9, 18, 23, 36, 42, 56, 72, 84, 86, 98, 99, 120, 138, 139, 145, 146], mainly involve applications in HIL/SIL simulations or driving simulators.

Vehicle models with LM-suspension were used or aimed to be used in driving simulators [25, 115, 136, 140], in virtual driver-in-the-loop simulations [103], and within model-in-the-loop, SIL, HIL and rapid control prototyping environments for the design of vehicle systems [14, 39] and electronic control units [101] as well as for vehicle controller development [110]. They were also a starting point for further model reduction and simplification processes [73, 75]. Vehicle models with LT-suspension were used in several HIL applications for testing of vehicle systems [8, 9, 99, 120, 138, 139], for controller development [72, 86], within a framework for distributed testing and verification of vehicles [84], as well as in several driving simulator applications for subjective evaluation of driving dynamics [5, 7, 36, 56]. Internal forces in the suspension, which are not accessible from an LT-suspension model, were calculated through coupling with a multibody kinematics model of a front suspension [138, 139] or with a black-box approach for load estimation [23]. Vehicle models with LM- or LT-suspension were also extended for simulating light or heavy commercial vehicles, including several model adaptations to take into account flexible chassis, rigid axles or different trailer-truck-combinations [8, 9, 98, 99, 101, 145, 146].

Vehicle models with LM- or LT-suspension are inherently real-time capable, as already discussed in Sect. 2.3.1. Due to their simple open-loop topology, the EOM can be easily formulated as ODE in independent coordinates, as in [25, 39, 72, 98, 99, 101, 103, 110, 115, 120]. Often, explicit integration methods, e.g., the explicit Euler method [101, 103] and Runge-Kutta methods of higher order [25, 110, 115, 120], but also the implicit single-step trapezoidal rule [98, 99], were used for integration of the EOM.

It should be noted that many of the publications did not provide detailed information about the formulation and integration of the EOM. This may be attributed to the ease of achieving real-time capability, the frequent use of commercial software tools, and a primary focus on practical applications rather than in-depth research on the models themselves. For this reason, they will not be further analyzed in the present paper, and the remainder will focus on MBS-suspension models.

4.2 Multibody suspension models

The analyzed publications related to MBS-suspension models focus on researching real-time multibody simulation methods. In the following, a detailed analysis will be presented, including EOM formulation and numerical integration methods, as well as model simplifications and approaches in the formulation, integration and implementation to achieve real-time capability. The section ends with an overview of those MBS-suspension models that include elastokinematic properties.

4.2.1 Formulation of the EOM

Global as well as topological formulations were employed to derive the EOM of the MBS-suspension models in the analyzed publications, as illustrated in Fig. 9. It should be noted that some publications used the same formulation, and some publications included more than one formulation. Each case was counted, so that the total number in Fig. 9 exceeds the number of publications.

Global and topological formulations are distributed in a ratio of about 1 : 2. It is noticeable that the relative proportion of elastokinematic MBS-suspension models is significantly higher among global formulations compared to topological ones. Overall, 55% of the formulations resulted in differential-algebraic EOM, with a larger relative share of global formulations. Ordinary differential EOM were mainly obtained with topological formulations. Most of the ODE, whether formulated globally or topologically, were obtained using coordinate reduction technique, transforming dependent coordinates into independent ones. The different formulations will be explained in the following.

Global formulation Cuadrado et al. proposed and extensively analyzed global formulations for kinematic MBS-suspensions [26, 27, 29–31, 40]. These formulations were used in a driving simulator and vehicle controller development [32, 78–80], and a vehicle model was validated against experimental data [100]. Global formulations were also utilized by Shiiba

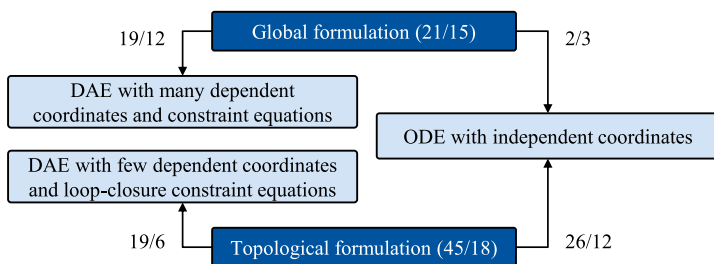


Fig. 9 EOM formulations in the analyzed publications (The numbers indicate their occurrence, with the first number related to kinematic and the second number related to elastokinematic MBS-suspension models)

et al. for several vehicle models with kinematic MBS-suspension, which were applied in a driving simulator [129–132] and a steering torque simulator [127, 128, 135]. Lúgrís et al. [71] extended a global formulation proposed by Cuadrado et al. for an MBS-suspension with flexible suspension bodies. A global formulation was applied for an MBS-suspension with bushings by Prescott et al. [24, 105, 108]. Their approaches were implemented in the commercial software packages LMS Virtual.Lab Motion and Simcenter 3D Motion [45, 109], and applied in combination with a vehicle control system [109]. A global formulation with a chassis-based kinematics description was proposed by Lee et al. [69] for an MBS-suspension with bushings. This was later extended by Kang et al. [55] to also incorporate a flexible suspension body. Hidalgo and García de Jalón [49] stated that incorporation of bushings into an MBS-suspension in a general fashion predetermines the use of global rather than topological formulations. Accordingly, they derived the EOM of a truck model with bushings with a global formulation. Carpinelli et al. [19] analyzed global formulations for MBS-suspensions with different number of bushings. A full vehicle model with flexible vehicle body was modeled with a global formulation by Schmitt and Seifried [123, 124].

Global formulations were also applied to MBS-suspensions without kinematic joints, resulting directly in ordinary differential EOM, in contrast to the cases of differential-algebraic EOM previously mentioned. Franco et al. [43] proposed a vehicle model based on the so-called discrete body dynamics method and used a global formulation to obtain the EOM. Here, all flexible suspension arms were replaced by spring-damper elements that connect the wheel mounts with the vehicle body, without additional joints. The authors of the present paper proposed an MBS-suspension model with extensive elastokinematic properties, whose EOM were also derived with a global formulation [2]. There, all joints were modeled flexible by means of bushing force elements.

Topological formulation Various topological formulation methods were found in the publications analyzed:

Semirecursive formulation with velocity transformation: Semirecursive formulation was first proposed by Jerkovsky [53] and has been extended by many authors. An introduction to the method is provided by Yu et al. [144]. Its basic idea is to first derive the differential-algebraic EOM of an open-loop system in Cartesian coordinates, and then transform it into ordinary-differential EOM with relative joint coordinates via velocity transformation. The involved velocity transformation matrix is calculated recursively along the kinematic chain by differentiating the functional relationship between Cartesian and joint coordinates. It only considers permissible directions of motions and eliminates the unknown constraint forces in the EOM. For closed-loop topologies, loop-closure constraints are added subsequently. Semirecursive algorithms combine the high computational efficiency of topological formulations with the simplicity of global formulations, making them widely used in multibody dynamics and implemented in commercial software packages. [83, 144]

Bae et al. [6] proposed a semirecursive formulation for a vehicle model with kinematic MBS-suspension, with loop-closure constraints formulated in the vehicle body reference frame. Lee et al. [68] applied a vehicle model based on the velocity transformation method in a driving simulator. Cuadrado et al. [26, 27, 30, 31, 40] proposed and extensively analyzed a method based on semirecursive formulation, which they called hybrid method. This method combines relative joint coordinates to describe the body kinematics with global coordinates to formulate the loop-closure constraints. It turned out to be more efficient and robust than the previously mentioned global formulations by Cuadrado et al. [26, 27, 29–31, 40]. The hybrid method was later analyzed in terms of parallel processing [49] and hardware acceleration methods [114]. Kim et al. [59, 61, 62] applied the semirecursive method within the

subsystem synthesis method and extended it by incorporating the compliance effect of bushings. Lugués et al. [71] extended a semirecursive formulation to account for the flexibility of suspension bodies.

Recursive formulation: Recursive algorithms formulate the EOM of open-loop topologies in relative coordinates along the kinematic chain. The calculation consists of three recursive steps [113]: In a first forward recursion, the kinematics of the bodies and the applied forces are calculated. In a second backward recursion, the generalized mass matrices and forces of the bodies are set up subsequently. In the last forward recursion, the generalized accelerations of the bodies are calculated one after another. Recursive algorithms are attractive for multibody systems consisting of large topological chains, because their computational effort increases only linearly with the number of coordinates. However, they may not be advantageous for vehicle models with a relatively low number of bodies because of computational overhead costs [13].

Cuadrado et al. [26, 29, 30] presented a recursive formulation for a kinematic MBS-suspension with closed-loop topology, ensuring loop-closure conditions with a penalty formulation. It turned out to be less efficient and, due to its case-dependent implementation, less attractive than the hybrid method. A recursive formulation was also employed for real-time vehicle simulation with kinematic MBS-suspension in the special-purpose software package SimCreator [102, 116], along with a graph-theory based method to convert a formulation in Cartesian coordinates into the recursive one. Burgermeister et al. [16] modeled an elastokinematic MBS-suspension featuring bushings with a recursive algorithm.

Jourdain's principle and Kane's method: The principle of virtual power, also known as Jourdain's principle, states that a constrained multibody system moves in a way such that the virtual power of the constraint forces and torques is equal to zero [104]. To obtain the EOM, the unknown constraint forces are formulated in terms of the inertia and applied forces. Virtual velocities are defined based on the constrained kinematics of the system. The EOM is then derived exploiting Jourdain's principle. Kane's method follows a similar approach utilizing auxiliary quantities called partial angular velocities and partial velocity vectors. An introduction to Kane's method is presented in [50]. Jourdain's principle and Kane's method are closely related as shown in [104].

Kane's equation with undetermined Lagrange multipliers was used by Choi et al. [20–22] to formulate the EOM of a full vehicle with kinematic MBS-suspensions considered as suspension composite joints. Rill [111] and Schmitt [123] utilized the principle of virtual power to obtain the EOM for vehicle models with elastokinematic properties. Furthermore, the EOM for an MBS-suspension model with bushings were formulated using Jourdain's principle in the commercial special-purpose simulation package Tesis DINA4 [17].

Further examples for topological formulations: Few further topological formulations were found in the analyzed publications. Masoudi et al. [73] formulated the differential-algebraic EOM of a vehicle model with kinematic MBS-suspension with a graph-theoretic approach. The model was subsequently reduced by means of proper orthogonal decomposition. Kracht et al. [34, 66, 70] developed a real-time object-oriented modeling method based on the Newton-Euler equations. The method assembles the vehicle model in a modular way as a composition of its individual elements and obtains the EOM for kinematic or elastokinematic MBS-suspensions as DAE or ODE. In the commercial software package SIMPACK and its extension SIMPACK Realtime, the EOM are formulated in a topological way using relative coordinates. It was used for detailed vehicle models with elastokinematic properties by several authors [8, 9, 77, 142].

Coordinate reduction in global and topological formulation Coordinate reduction was performed in the analyzed papers in combination with global EOM formulations. Serban and

Haug [126] proposed a coordinate reduction to globally independent coordinates for a global formulation of a vehicle model with kinematic MBS-suspension. They showed that coordinates related to the suspension's compliant elements, such as the spring or shock absorber, and coordinates related to its control elements, such as the steering rack, can be selected as globally independent coordinates that are valid for the entire motion range of the suspension. A similar idea was followed by Kim and Wang [60] for a global formulation within the subsystem synthesis method. They introduced new independent coordinates in addition to the Cartesian coordinates that remain valid during the entire simulation. Additional constraint equations related to the new coordinates were added, and the extended constraint Jacobian was partitioned to express the dependent Cartesian coordinates in terms of the independent coordinates. Carpinelli et al. [19] investigated potential efficiency improvements through coordinate reduction in a global formulation of elastokinematic MBS-suspensions. They applied automatic coordinate switching and obtained efficiency improvements for an MBS-suspension model with a high number of ideal joints and constraints. In contrast, coordinate switching proved to be less efficient for an MBS-suspension model with a high number of bushings instead of ideal joints.

Various authors proposed and analyzed coordinate reduction within topological formulations. Cuadrado et al. [26, 27, 30, 31, 40] analyzed a topological semirecursive formulation with subsequent coordinate reduction for a vehicle model with kinematic MBS-suspension. It proved to be similar efficient to the hybrid method mentioned before, but it showed drawbacks in terms of numerical robustness and limited range of validity of the independent coordinates. A similar method was presented by García de Jalón et al. [44], which was called double-step semirecursive formulation. In the first step of this formulation, the differential-algebraic EOM of a closed-loop topology is obtained in relative joint coordinates using semirecursive formulation. To this end, closed-loops are opened by means of the joint-cut or the rod-removal technique, and loop-closure constraints are formulated in relative coordinates. In the second step, coordinate partitioning based on Gaussian elimination and full pivoting transforms the EOM into ODE. This approach was applied for real-time full vehicle simulation of a bus with kinematic MBS-suspension [44]. The implementation of this method was further improved to reduce its computation times by Hidalgo and García de Jalón [49]. The double-step semirecursive formulation was later extended, analyzed and applied in many publications. Pan et al. compared the efficiency of different possibilities to open closed-loop topologies [93], and validated the method against results obtained with a commercial general-purpose software package [94]. Various integration methods for solving the ODE were proposed and compared in [87, 91, 92, 94]. Dai et al. [33] extended the method with a fork-arm removal technique that allows to open the closed-loop topology removing fork-arms. It was extensively applied for developing control strategies [47] and data acquisition for training of various machine learning models [81, 82, 95, 96, 134]. All mentioned examples of the double-step semirecursive formulation only considered purely kinematic MBS-suspensions.

Kim and Wang [60] combined a semirecursive formulation with the generalized coordinate partitioning technique within the subsystem synthesis method. This approach was later extended by Kim and Jeong [58] with the approximate function method. They manually selected the translational joint coordinate in the spring-damper unit of the suspension as the independent coordinate. Rather than deriving the transformation between dependent and independent coordinates from the constraint Jacobian, they described it through polynomial functions that approximate results from quasistatic wheel travel simulations. Kim et al. [61] extended the combination of semirecursive formulation and generalized coordinate partitioning by a quasistatic consideration of the bushing compliance effect in an elastokinematic MBS-suspension.

Pankiewicz and Rulka [97, 118] presented the macro-joint approach, which employs coordinate partitioning to address complex suspension kinematics. The approach considers the suspension as a macro-joint that is internally described by dependent joint coordinates. Through coordinate partitioning, the kinematics of the macro-joint is expressed by independent coordinates, and the resulting EOM are ODE. The method is applicable to both kinematic and elastokinematic MBS-suspensions with bushing compliance.

Embedding technique, which is a symbolic counterpart to numerical generalized coordinate partitioning, was used by Uchida and McPhee [140, 141]. Kinematic MBS-suspensions were described with a mixed set of absolute and relative joint coordinates. Appropriate independent coordinates were manually selected, and the embedding technique was employed to symbolically derive the ordinary differential EOM.

Further examples for coordinate reduction with coordinate partitioning for kinematic MBS-suspensions can be found in [37, 67]. Furthermore, real-time simulation of vehicle models with elastokinematic MBS-suspensions featuring bushings in the commercial special-purpose software package IPG CarMaker [48, 122] is based on MESA VERDE [143], which utilizes the principle of virtual power and coordinate partitioning of relative joint coordinates.

4.2.2 Integration of the EOM

A variety of different integration methods were investigated and applied for real-time simulation in the analyzed publications. Figure 10 provides an overview of the forms of the EOM, i.e., ODE or (transformed) DAE, and the integration methods. Their occurrence was counted as follows: A solution approach was defined as the combination of the EOM formulation method (see Sect. 4.2.1), the form of the EOM, and the integration method. For each solution approach identified in the analyzed publications, the corresponding numbers in Fig. 10 were increased by one. Some solution approaches appeared in multiple publications, increasing the corresponding numbers with each appearance. The connecting lines in Fig. 10 represent the number of solution approaches that relate a specific EOM form with a specific integration method. Among the forms of the EOM, ordinary differential EOM account for 45% of the solution approaches, as found in 43 publications [2, 8, 9, 17, 19, 26, 27, 30, 31, 33, 37, 40, 43, 44, 47–49, 58, 60, 61, 66, 67, 70, 77, 81, 82, 87, 91–97, 111, 118, 122, 126, 133, 134, 140–142]. About 52% of the solution approaches pertain to differential-algebraic EOM, most of which were transformed prior to numerical time integration. They are presented in more detail in the following subsection. Regarding the integration methods, most of the counted solution approaches utilized explicit or implicit methods with almost similar portions of 43% and 38%, respectively. Linear-implicit methods were observed infrequently, with a share of 7% of the solution approaches. It should be noted that some publications did not provide specific information about the EOM form and/or integration method.

Figure 11 shows the integration step sizes used for real-time simulation with the different solution approaches, categorized into four groups. The statistics only includes step sizes that were explicitly reported in the analyzed publications, with some publications also examining different step sizes. Interestingly, step sizes larger than 1 ms were often used, although this is a typical value for real-time environments in the field of vehicle dynamics simulation (cf. Sect. 2.2). It is also noteworthy that slightly more than one-third of the publications did not specify the step size used, despite its direct impact on real-time execution. Further discussion on the step sizes follows in the subsequent subsections.

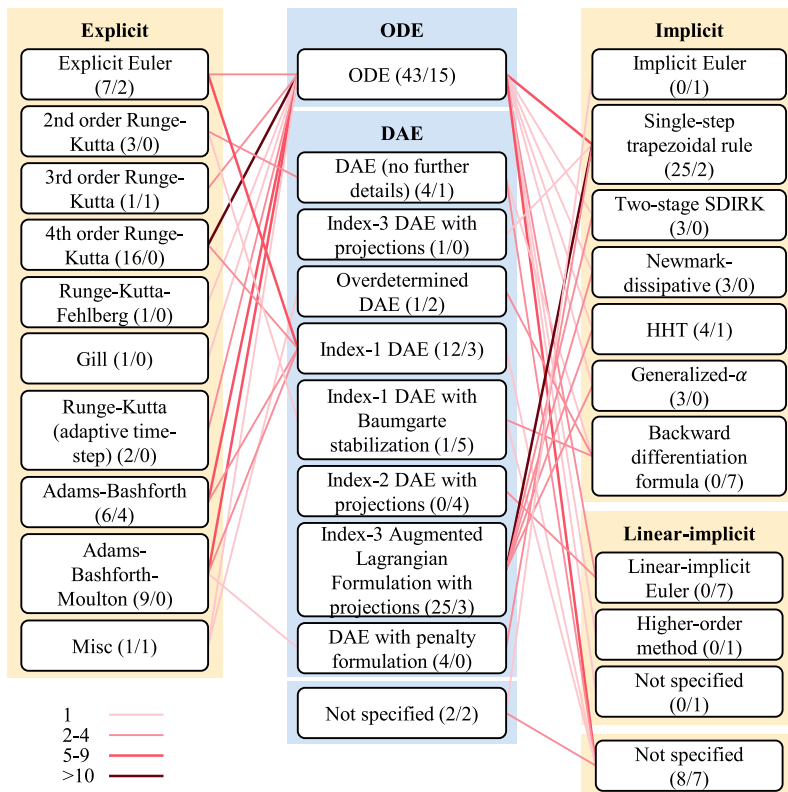


Fig. 10 Solution approaches in the analyzed publications as combination of the EOM formulation, form of the EOM and integration method (The numbers indicate their occurrence, with the first number related to kinematic and the second number related to elastokinematic MBS-suspension models)

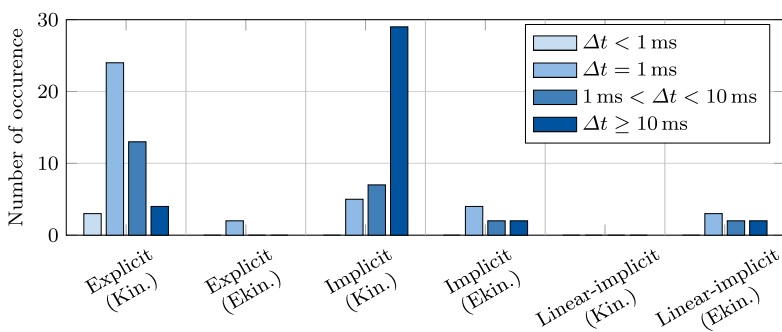


Fig. 11 Integration step sizes Δt of the solution approaches in the analyzed publications, sorted by integration methods (Kin.: kinematic MBS-suspension models; Ekin.: elastokinematic MBS-suspension models)

Transformation of differential-algebraic EOM Direct integration of the index-3 DAE, or integration of the overdetermined DAE with the constraints on the position, velocity and acceleration level were only rarely pursued in the analyzed publications [6, 29, 55, 69].

Often, the DAE was transformed analytically and integrated in index-1 form [20–22, 59, 61, 62, 127–132, 135]. In addition, Baumgarte stabilization was applied in some publications to prevent the drift-off related to index-reduced DAE [24, 45, 102, 105, 108, 109]. Alternatively, a non-iterative projection method that projects the numerical solution on the constraint manifold was proposed for real-time simulations by Arnold et al. [3], and later applied for vehicle dynamics simulation in [19, 123, 124].

An index-3 augmented Lagrangian form of the EOM was used in the global formulations and the hybrid method by Cuadrado et al. [26, 27, 29–32, 40, 49, 78–80, 100, 114], in a global and a semirecursive formulation with flexible bodies by Lugić et al. [71], and a global formulation with bushings by Hidalgo and García de Jalón [49]. Penalty-based schemes were also used by Takouda et al. [135] and by Cuadrado et al. [26, 29, 30] in fully-recursive formulations.

Explicit integration methods The explicit Euler method as the simplest explicit integration method was used for real-time vehicle dynamics simulation in numerous cases [20–22, 97, 118, 130, 131, 140, 141]. It is based on forward difference approximation and requires the EOM to be evaluated only in the current time step [4].

Higher-order explicit Runge-Kutta methods exploit more information from the EOM through additional function evaluations in intermediate stages between the current and the next time step [4]. They require more computational effort per integration step, but their higher order of convergence results in a more accurate solution. Runge-Kutta methods with different number of stages and order were applied in the analyzed publications. This includes the 2nd order Runge-Kutta method [34, 102, 116] and the 3rd order Bogacki-Shampine method [66, 70]. In most cases, however, the 4th order Runge-Kutta method was used [33, 44, 47, 49, 82, 87, 91–96, 129–131, 134]. Furthermore, the Gill and the Runge-Kutta-Fehlberg method were applied in full vehicle simulation based on the topological double-step semirecursive formulation, but they did not provide benefits compared to the 4th order Runge-Kutta method [94]. Pan et al. [92] also introduced an adaptive time-step Runge-Kutta algorithm with error control in combination with the double-step semirecursive formulation. The algorithm adaptively increases the integration step size and proved to be more efficient than the fixed-time step 4th order Runge-Kutta method, achieving simulations much faster than real-time [92, 94]. However, its usability in real-time applications is limited, as they typically require fixed-step size algorithms.

An alternative to single-step Runge-Kutta methods are linear multi-step methods. They reuse function values of the EOM already known from previous time steps to improve the accuracy of the solution, rather than evaluating the EOM in additional intermediate stages [4]. The explicit Adams method, also called Adams-Bashforth method, was applied in many publications [58–62, 67, 126, 127], as was the Adams-Bashforth-Moulton method based on a predictor-corrector scheme [82, 91, 92, 94, 96, 132, 134, 135]. Because the Adams-Bashforth-Moulton method requires only two instead of four function evaluations per integration step, it proved to be more efficient than the 4th order Runge-Kutta method, nevertheless providing results with similar accuracy [91, 92, 94].

Further explicit integration methods proposed and applied in the analyzed publications are a centered finite difference procedure [43] and a tailored integration method for over-determined DAE [6].

Explicit integration methods were merely used for kinematic MBS-suspension models, as the statistics in Fig. 10 shows. Most often, step sizes of 1 ms [6, 20, 22, 33, 44, 49, 58, 66, 82, 87, 91–94, 96, 132, 140, 141] or 2 ms [67, 116, 127, 129, 130, 135] were selected. Higher step sizes, in the range of 10 ms and more, were reported less frequently, see Fig. 11

[34, 60, 82, 126]. In some cases, real-time simulation was also performed with a step size of 0.5 ms [87, 92, 130]. Some of the aforementioned publications include simulations of vehicles with elastokinematic MBS-suspensions [43, 59, 61, 62, 70, 97, 118, 123]. Due to the limited stability properties of explicit integration methods, model simplifications were necessary to avoid high numerical stiffness of the EOM and enable stable simulation with step sizes in the range of milliseconds. These simplifications will be described in detail in Sect. 4.2.3.

Implicit integration methods The simplest implicit integration method, which is the implicit Euler method, was only once applied in the analyzed publications, in combination with a symbolic manipulating method called inline integration [63]. It relies, in contrast to the explicit Euler method, on a backward difference approximation [4]. The implicit single-step trapezoidal rule is a second-order method that approximates the state in the next time step with a trapezoidal scheme. It was extensively used in combination with various EOM formulations for full vehicles with kinematic MBS-suspension by Cuadrado et al. [26, 27, 29–32, 40, 78–80, 100, 114]. A special version of implicit Runge-Kutta methods with greater simplicity and efficiency are the so-called singly diagonally implicit Runge-Kutta (SDIRK) algorithms. A two-stage SDIRK method was analyzed by Dopico et al. [40] and was found to be inferior to the implicit single-step trapezoidal rule in terms of accuracy, robustness and computational effort. The Newmark dissipative integrator, the HHT integrator, and the generalized- α integrator belong to the family of structural integrators. They were also compared to the implicit single-step trapezoidal rule by Cuadrado et al. [31], and produced better results for large multibody systems, especially the Newmark dissipative integrator. The HHT integrator was later applied by Hidalgo and García de Jalón [49] in combination with a semirecursive formulation. Another generalization of the implicit Euler method to higher order methods is the backward differentiation formulae [4], which use already known solutions from previous time steps for increased accuracy. They are especially suited for numerically stiff problems [4] and were often applied for real-time simulation of vehicle models with elastokinematic MBS-suspension [16, 24, 55, 69, 105, 108, 109]. The implicit Euler method [63], the implicit single-step trapezoidal rule [71] and the HHT integrator [49] were also used for vehicles with elastokinematic properties.

The good stability properties of implicit integration methods allowed to use larger integration step sizes to compensate for their higher computational costs per integration step and achieve faster simulation. In fact, the trapezoidal rule, the structural integrators and the 2-stage SDIRK integrator were used with integration step sizes of the order of tens of milliseconds for vehicle models with kinematic MBS-suspension [26, 27, 29–31, 40, 71, 100, 114], see Fig. 11. However, vehicle models with high numerical stiffness, in particular with elastokinematic MBS-suspension, were integrated with smaller step sizes, e.g., 1 ms or 2 ms [24, 63, 69, 105, 108, 109].

To circumvent the problem of varying computational costs per time step, which can restrict the applicability of implicit integration methods for real-time simulations, the number of iterations for solving the involved nonlinear equation systems can be limited. This approach makes the integration scheme deterministic and ensures a constant, a priori known computational effort per time step. It was proposed by Prescott et al. [24, 105, 108] for complex vehicle models with elastokinematic properties and was also used by other authors [45, 109]. A fixed number of three iterations per time step of a 2nd order backward differentiation formula with a step size of 1 ms has proven to be a good choice for real-time integration of such vehicle models [105].

Linear-implicit integration methods In addition to common explicit and implicit integration methods, the analyzed publications also investigated linear-implicit methods. They are based on implicit integration schemes but do not require iterative solution methods as they exploit a linearization of the EOM [3]. They are especially attractive for real-time integration of numerically stiff EOM, as they combine a comparatively low computational effort, constant in each time step, with excellent stability properties. However, they were found only infrequently in the analyzed publications.

Rill [111] demonstrated how the implicit Euler method can be modified into a linear-implicit integration scheme. Starting from a semi-implicit Euler scheme, the vector of generalized forces in the unknown next time step is expanded into a Taylor series neglecting higher order terms. Through this linearization, the integration only requires the non-iterative solution of a linear equation system, while still being stable. This approach can also be interpreted as the implicit Euler method with only one iteration of the Newton method to solve the nonlinear equation system [3]. The Jacobians of the generalized forces have to approximate all stiff terms to achieve unconditional stability [3]. Rill [111] proposed to calculate them analytically in each integration step, neglecting the partial derivatives of the inertia forces. He demonstrated the efficiency of the method in a real-time simulation of a quarter vehicle including stiff bushings and a dynamic damper model with dry friction. Arnold et al. [3] extended the linear-implicit Euler method to DAE and proposed an index-2 formulation in combination with non-iterative projection to avoid the drift-off effect. Schmitt and Seifried [123, 124] adapted this approach for a full vehicle model with a flexible vehicle body, exploiting analytical Jacobians and a comparatively coarse step size of 10 ms or higher. Carpinelli et al. [19] investigated efficiency improvements in the linear-implicit Euler method by switching from the index-2 DAE formulation with non-iterative projection to an ODE formulation by means of coordinate reduction. The linear-implicit Euler method may have limited accuracy in complex applications, cf. [3, 19]. Consequently, a higher-order multi-stage method was adapted by the authors of the present paper for real-time simulation of complex MBS-suspensions with extensive elastokinematic properties [2]. They proposed to not repeat the analytical Jacobian calculation in each integration step but to perform a number of integration steps before updating the Jacobians. Efficiency was further improved by exploiting the linearization to approximate the generalized forces in the intermediate stages. Further applications of linear-implicit integration can be found in [17, 142]. As Fig. 11 illustrates, step sizes used with linear-implicit integration methods are 1 ms [2, 19, 142], 2 ms [2, 111] and higher [123, 124].

4.2.3 Simplifications in the system modeling

The modeling of an MBS-suspension model includes the definition and parameterization of the involved bodies, joints and force elements. This not only determines the size, topology and complexity of the model, but also influences the formulation of the EOM, its numerical properties and its numerical integration. In the analyzed publications, different approaches in the modeling were identified that aim at reducing the computation times. They are usually associated with a modification or simplification of the model and affect its accuracy. Figure 12 provides an overview of the identified approaches and their occurrence in the analyzed publications. The approaches are explained in more detail in the following subsections.

Massless link approach The massless link approach assumes that mass and moment of inertia of a suspension link are relatively small and can be neglected. Consequently, the link is removed and replaced by a distance constraint. This reduces the model size, because

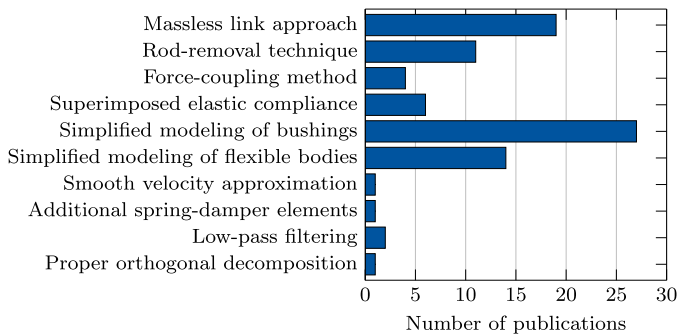


Fig. 12 Simplifications in the system modeling to reduce computation times in the analyzed publications

the coordinates and constraint equations associated with the removed body and the related joints are eliminated from the EOM and replaced by only one distance constraint equation [68, 116]. Although this approach retains the exact kinematics of the suspension [20], it is connected with a significant model simplification. Mass and moment of inertia of the replaced link are either completely neglected or distributed among the neighboring components [123]. Furthermore, problems arise when forces are applied on the suspension links, e.g., from the suspension's shock absorber [58].

The massless link approach was applied by several authors in both global and topological formulations. Often, only tie rods [60, 88, 89, 102, 116, 133] or some additional links [6, 68, 119, 128, 132, 135] were considered as massless links. Choi et al. [20–22] proposed to model kinematic MBS-suspensions as suspension composite joints, in which all links with relatively small masses are treated as massless. Similarly, Pankiewicz and Rulka [97, 118] presented an automatic component-oriented model reduction based on massless links, called the macro-joint approach. This approach results in ordinary differential EOM and can also handle the stiffness properties of bushings located between the vehicle body and the suspension links. More recently, Schmitt and Seifried [123, 124] presented a real-time capable vehicle model with a flexible vehicle body. In this model, all suspension bodies except for the wheel mount and the wheel are neglected, meaning that the wheel mount is guided solely by constraints.

Rod-removal technique The rod-removal technique is a method for opening closed-loop topologies, as an alternative to the joint-cutting method [44]. Here, a suspension link is simplified as a slender rigid body with uniformly distributed mass and negligible moment of inertia about its longitudinal axis, connected to its neighboring bodies through spherical joints [90]. This suspension rod is replaced by a distance constraint in order to reduce the model size; however, unlike the massless link approach, its dynamic properties including inertia forces and external forces are still entirely included [90].

The rod-removal technique was considered by García de Jalón et al. [44] in the double-step semirecursive formulation. Pan et al. analyzed and extended the technique in several publications: They compared various possibilities of joint-cuts and rod-removals on the example of a sedan vehicle [93], validated the approach by comparison with a commercial general-purpose software [94], and employed different integration methods to solve the EOM [92, 94]. Their findings indicated that higher efficiency is achieved by opening closed-loop systems in a way that generates less branches, loop-closure constraints and bodies, and that the rod-removal technique leads to slightly less accurate results due to the approximation of a uniform mass distribution. An extension of the rod-removal technique to fork arms

was presented by Dai et al. [33], where a fork arm is replaced by two distance constraints. Vehicle models utilizing the rod-removal technique were subsequently applied in several applications for vehicle controller development or data acquisition for machine learning [47, 81, 82, 95, 96, 134]. The rod-removal technique was only applied to kinematic MBS-suspension models in the analyzed publications.

Force-coupling method The force-coupling method replaces ideal joints with force elements, enforcing the coupling between bodies through applied forces rather than constraint forces with unknown Lagrange multipliers [121]. This approach eliminates constraint equations and can ultimately convert differential-algebraic into ordinary differential EOM, preventing the difficulties of real-time integration of DAE. However, the stiffness of the force elements must be sufficiently high to replicate the behavior of ideal joints. This can increase the numerical stiffness of the EOM and thus cause stability problems in the integration.

Kracht and Schramm [66] replaced ideal joints with bushing force elements in order to open closed-loop topologies and avoid iterative methods for solving their kinematics. They compared several bushing models and optimized their stiffness and damping parameters to achieve accurate and stable real-time simulation with an explicit integration method. Bushing force elements were also used in vehicles modeled in the commercial software package SIMPACK, both to avoid constraint equations and to represent the elastokinematics of the suspension [77, 142]. Linear-implicit integration schemes were used for stable real-time simulation. The authors of the present paper made extensive use of bushings to model the elastokinematics of suspensions [2]. To directly obtain the ordinary differential EOM from a global formulation, they exploited force coupling and replaced the few joints in the suspensions, such as the translational joints in the spring-damper units, with appropriately parametrized bushings elements. The numerically stiff ODE were integrated with a higher-order linear-implicit integrator.

Superimposed elastic compliance Superimposing elastic compliance means to overlay the motion of kinematic MBS-suspension models with elastokinematic effects. The compliance of the suspension is approximated in a quasistatic way, which reduces model complexity and avoids high numerical stiffness of the EOM and associated stability problems in the numerical integration. However, the internal dynamics within the suspension is strongly simplified, diminishing model accuracy in the higher frequency range.

In a tractor-trailer model presented by Salaani et al. [119], compliant steer effects were superimposed on the kinematic steering in a quasistatic way. The compliant steering angle was approximated based on the overall suspension stiffness, which was determined through experiments with a real vehicle. Similarly, Pankiewicz and Rulka [97, 118] proposed in their macro-joint approach to incorporate the overall suspension compliance between wheel mount and vehicle body in a quasistatic way by superimposing elastokinematic wheel alignment changes on the kinematically guided wheel motion. In contrast, Kim et al. [59, 61, 62] considered the compliance effect of individual bushings in the suspension. They employed a kinematic MBS-suspension model to simulate the motion of the suspension bodies and obtain reaction forces in selected joints. In each integration step, the hard points of this kinematic MBS-suspension were shifted according to corresponding quasistatic bushing deformations obtained from the joint reaction forces. This approach could approximate the elastokinematic wheel motion of a suspension, but with deviations even in quasistatic wheel travel simulations.

Simplified modeling of bushings Bushing force elements are used to model the elastokinematic behavior of suspensions resulting from local compliance. Despite the availability of complex bushing models (see Sect. 2.3.2), bushings are often simplified for real-time simulations to reduce model complexity and computation times. It should also be noted that many publications did not provide detailed information about the used bushing models [8, 9, 16, 24, 45, 48, 49, 63, 77, 105, 108, 109, 122, 142].

The most complex bushing models about which information was provided consider linear or nonlinear stiffness and damping properties in all six translational and rotational directions, as used by Lefèvre et al. [70] and the authors of the present paper [2]. A bushing model with nonlinear stiffness and damping properties but only in two translational directions in the plane was considered by Rill [111]. Carpinelli et al. [19] employed a spatial model with linear stiffness and damping properties. More simplified bushing models consider only linear stiffness properties in translational directions, e.g., within the subsystem synthesis method by Kim et al. [59, 61, 62] and the chassis-based kinematic formulation by Lee et al. [55, 69]. Other approaches consider linear bushing stiffness properties only in the direction of the connected suspension links, e.g., the macro-joint approach by Pankiewicz and Rulka [97, 118]. Similarly, Butz et al. [17] proposed an optimization-based parameter estimation method to find the stiffness values of bushings in the direction of the respective control arms. In the discrete body dynamics approach proposed by Franco et al. [43], the compliance of bushings is added to the translational spring-damper elements that represent the flexible control arms.

Simplified modeling of flexible bodies Bodies are modeled as flexible in order to represent elastokinematic effects resulting from their structural compliance, including the vehicle body and the suspension bodies. In the analyzed publications, different approaches of simplified modeling for reduced computation times were found.

The lumped mass approach for flexible bodies divides a flexible body into a finite number of rigid bodies connected through force elements that are parameterized to model the body's deformation behavior in a simplified way. For example, Salaani et al. [119] divided a truck chassis into two bodies with a torsional spring in between to model the twisting about its longitudinal axis. Prescott et al. [108] proposed to divide the vehicle body into a front and a rear part and connect them with a bushing force element. Although this modeling was primarily aimed to enable parallel processing, the parameters of the bushing can be tuned to the behavior of a flexible vehicle body. This approach was subsequently considered in other publications [24, 45, 105].

The discrete body dynamics approach is based on the principles of the discrete element method, which models granular materials as particles connected by force elements. It was proposed for vehicle dynamics simulation by Franco et al. [43]. They considered the vehicle body and the wheel mounts as particles that are connected by massless spring-damper elements. The stiffness and damping properties of these force elements were chosen to approximate the flexible behavior of the replaced suspension rods.

More detailed information about the deformation behavior of flexible components is provided by modal representation based on analytical or finite element (FE) models. A vehicle model with a flexible vehicle body based on a detailed FE model was presented by Schmitt and Seifried [123, 124]. They modeled the vehicle body through a total of 40 modes obtained through Craig-Bampton model order reduction. The suspension was strongly simplified using the massless link approach. Venrooij et al. [142] simplified a detailed vehicle model with a flexible vehicle body implemented in the commercial software package SIMPACK. The number of flexible modes determined through condensation was reduced to achieve real-time capability. This decreased accuracy, but the results obtained were still better than those

of a rigid body model, particularly in the frequency range above 20 Hz. Similar model simplifications were found in [8, 9, 77]. A full vehicle model with flexible suspension bodies was presented by Lugić et al. [71]. They considered 12 bodies in the full vehicle as flexible with a total of 64 modes determined through Craig-Bampton reduction method. Different formulations of the EOM and the use of inertia shape integrals were analyzed in terms of computation times. Kang et al. [55] extended their chassis-based kinematic formulation to account for body flexibility, presenting a full vehicle with a coupled torsion beam axle, where the torsion beam was described through modal representation.

It should be noted that modeling elastokinematics as overall suspension compliance based on virtual or experimental measurements also accounts for body flexibility without explicitly modeling flexible bodies [97, 118, 119].

Smooth velocity approximation Smooth velocity approximation is an analytical model reduction method based on a quasistatic approach that aims to reduce the numerical stiffness of a multibody system by smoothing out the velocities of fast-moving small masses. Burgermeister et al. [16] proposed this method for suspension systems that contain small-mass bodies connected through stiff force elements such as bushings. Here, the bodies in the system were separated into slowly- and fast-moving bodies, taking into account masses, acting forces and topology of the system. High-frequency components in the solution of the EOM, associated with the fast-moving small-mass bodies, were prevented by neglecting their inertia forces, without completely removing the bodies. This strategy allowed higher step sizes of the numerical solver and reduced computation times of a full vehicle by up to 50%. However, deviations in the solution occurred if the neglected inertia was too high.

Introduction of additional spring and damper elements The introduction of additional spring and damper elements aims to reduce the numerical stiffness of MBS-suspension models with elastokinematic properties [70]. In an MBS-suspension modeled with the real-time object-oriented modeling method by Kracht, additional springs and dampers were positioned in series to the actual bushing elements [70]. This reduced the overall stiffness and enabled usage of explicit integration methods, without altering the forces in the bushings. The additional stiffness and damping parameters were optimized for selected load cases to achieve stable simulation and limit the errors introduced by this model modification.

Low-pass filtering Introducing low-pass filters into MBS-suspension models modifies the numerical properties of their EOM and enables stable simulation with explicit integration schemes. Kracht et al. [66, 70] proposed to filter reaction or applied forces to achieve a PT1-behavior of their build up, equivalent to a dynamic friction effect. This helped to avoid algebraic loops and transform differential-algebraic into ordinary differential EOM, and to reduce the numerical stiffness of the system. The low-pass filters were parametrized depending on the application and the desired frequency range, which was chosen to cover frequencies up to 30 Hz.

Proper orthogonal decomposition Proper orthogonal decomposition is a projection scheme for model order reduction of nonlinear systems. It projects the states of the original system onto a smaller subspace whose basis is constructed from snapshots obtained from experiments or simulations with high-fidelity models. Masoudi et al. [73] applied this method to reduce the size of a vehicle model with kinematic MBS-suspension for real-time simulation. High accuracy was achieved for driving maneuvers with the same inputs as those used to obtain the snapshots, with simulation times 23% faster than the original model. However, model validity may be limited, as deviations can occur for perturbed inputs [73].

4.2.4 Approaches in the EOM formulation, EOM integration and implementation

In Sect. 4.2.1 and 4.2.2, it was shown that various EOM formulation and integration methods were employed for real-time vehicle dynamics simulation with MBS-suspension models. Within these methods and their implementation, several approaches were identified to further reduce computation times. These approaches were divided into different categories, which are shown together with their occurrence in Fig. 13. They are explained in more detail below, noting that some approaches fall into more than one category.

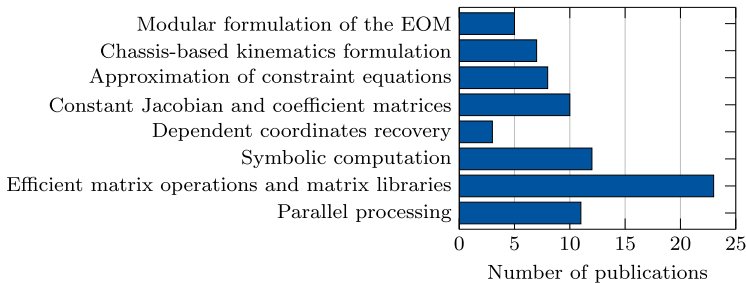


Fig. 13 Approaches in the EOM formulation, integration and the implementation to reduce computation times in the analyzed publications

Modular formulation of the EOM Modular formulation of the EOM aims to divide a multi-body system into several subsystems with virtual reference bodies that are analyzed independently. Kim [57] proposed the subsystem synthesis method for vehicle system dynamics, which has been further developed for real-time simulations. It considers the vehicle body as a base body and models each suspension as a subsystem. Consequently, the vehicle's dynamics is not described through a single system of EOM for the complete vehicle but through several smaller systems of EOM for the base body and each subsystem, which are solved more efficiently [58]. To account for the kinematic coupling between the vehicle body and the suspensions, effective masses and forces of the suspensions are considered in the analysis of the vehicle body, and the vehicle body serves as a virtual reference body for the suspension subsystems. Due to the modular structure of the method, the EOM of the subsystems can be generated with different formulations. Kim and Wang [60] developed two different subsystem synthesis methods with independent coordinates: a global relative Cartesian coordinates formulation with coordinate partitioning, and a topological relative joint coordinates formulation with velocity transformation and coordinate partitioning. The latter was later extended with the approximate function approach [58]. Furthermore, elastokinematic MBS-suspensions were modeled by means of superimposed elastic compliance [59, 61, 62]. The subsystem synthesis method is also suited to implement parallel processing [54], but this was not considered for road vehicles in the analyzed publications.

Chassis-based kinematic formulations Chassis-based formulations describe the suspension kinematics relative to the vehicle body, freeing the kinematic description from the large vehicle body motion. Then, the kinematic calculations can be further simplified by assuming small relative motions of the suspension bodies. For example, velocity transformation matrices, constraint Jacobians, and related coefficient matrices of linear equation systems

can be approximated as constant. They need to be computed and decomposed only once in the initial configuration, significantly reducing the computational costs during real-time simulation.

Bae et al. [6] proposed a chassis-based kinematic representation within a topological semirecursive EOM formulation. The involved velocity transformation matrix and constraint Jacobian were assumed to be constant, and position, velocity and acceleration analyses were performed using the Newton chord method with an LU-decomposition from the initial configuration. Shiiba et al. [129–132] introduced a global formulation where the suspension kinematics is described relative to the vehicle body. The involved constraint Jacobian was approximated to remain constant, resulting in a constant coefficient matrix of the index-1 differential-algebraic EOM. Chassis-based kinematic formulations for elastokinematic MBS-suspensions were presented by Lee et al. [69], considering bushings, and later by Kang et al. [55], additionally incorporating a flexible suspension body. They utilized a global EOM formulation with constant Jacobians of the constraint equations and generalized forces.

Approximation of constraint equations The approximation of constraint equations aims to simplify the kinematic analysis and reduce the computational effort for real-time simulation. Essentially, assuming constraint Jacobians to be constant, as done in chassis-based kinematic formulations, is related to a linearization of the constraint equations, with the benefits already described in the previous section. A different approximation, based on a polynomial representation, was proposed by Kim and Jeong [58], called the approximate function approach. They formulated the EOM of a kinematic MBS-suspension in dependent joint coordinates and further reduced it through coordinate reduction, selecting the displacement of the spring-damper unit as the independent coordinate. The relation between dependent and independent coordinates was approximated through polynomial functions that fit results from quasistatic wheel travel simulations. Compared to classical coordinate partitioning with evaluation of the constraint equations, the computational complexity of real-time simulations could be reduced, but the accuracy of the results depended on the selected polynomial order.

Constant Jacobian and coefficient matrix approach Calculation of Jacobians and related coefficient matrices of linear equation systems, as well as their decomposition, often belong to the most time-consuming processing steps of a multibody simulation. To prevent this computational burden, the constant Jacobian and coefficient matrix approach aims to reuse matrices that have already been computed. The approaches presented in the previous section that assume constant constraint Jacobians fall into this category. Other approaches consider the Jacobians of the generalized forces, also known as stiffness and damping matrices. The authors of the present paper implemented a linear-implicit integration scheme with analytical Jacobians of the generalized forces for real-time simulation of elastokinematic MBS-suspensions [2]. The implementation allowed to reuse the Jacobians for a few integration steps and update them at larger intervals, reducing the need for expensive matrix computation and decomposition. However, the size of the Jacobian update interval was limited because of the high nonlinearity of the considered suspension systems. Hidalgo and García de Jalón [49] implemented an implicit HHT integration scheme with approximated tangent matrix, in which the system's stiffness and damping matrices from the last iteration of an integration step were reused in the first iteration of the next integration step. Pan et al. [87] improved the efficiency of an explicit Runge-Kutta algorithm with an iterative refinement process to solve the associated linear equation systems, reusing the decomposed coefficient matrix from previous integrator stages.

Coordinates recovery If the EOM of a multibody system are formulated in a subset of dependent or independent coordinates, it may be necessary to recover unknown dependent coordinates, e.g., to obtain complete information about the motion of the bodies or to calculate the forces in force elements. To avoid iterative solution procedures, dependent coordinates are typically recovered through numerical integration of their derivatives that are obtained from the constraint equations at the velocity or acceleration level, cf. equations (3) and (4). Examples from the analyzed publications include [6, 60, 87, 91, 92, 94]. This solution strategy was also combined with Baumgarte stabilization or projection schemes to prevent the drift-off from the constraint manifold [19, 97, 118]. Alternatively, the dependent coordinates can be recovered directly from the constraint equations at the position level. This usually requires the time-expensive iterative solution of nonlinear equation systems.

Some approaches for efficient coordinates recovery at the position level were found in the analyzed publications. Serban and Haug [126] decoupled the large set of constraint equations related to Cartesian coordinates into several smaller sets, which were solved in a computationally efficient sequence. The decoupled subsets were identified from the vehicle system's topology using a weighted graph representation. Uchida and McPhee [140, 141] proposed a triangularization of the constraint equations. They simplified and symbolically reduced a subset of the constraint equations to a block-triangular form, which allowed to recover the dependent coordinates in an efficient recursive way. This process was performed manually as well as in a more automated manner utilizing Gröbner bases.

Symbolic computation Symbolic computation is a method of generating the analytical form of equations by manipulating arithmetic operators and strings of alphanumeric characters. It combines the advantage of manual analytical modeling, which can produce optimized mathematical equations, with the advantage of numerical modeling in terms of a high level of generality. Highly efficient equations are generated, e.g., by simplifying arithmetical operations and trigonometric expressions, identifying and removing unnecessary calculations, and making more efficient use of computer memory. Symbolic generation is performed once in the model generation phase and produces highly portable computer code that can be exported to other model environments for use in (real-time) simulations. [38]

Lee and Lee [67] developed the symbolic computation package AUTOCODE, based on the symbolic computing environment Maple, and generated simulation code for a full vehicle model with kinematic MBS-suspension. The code was automatically generated based on the model topology, body properties, joint geometries and a few other parameters. Symbolic expressions were manipulated, and repetitive calculations were substituted with intermediate variables. Complex force elements were incorporated as subroutines. While the system's EOM were obtained symbolically, the final mass matrix inversion was performed numerically to take advantage of efficient numerical computation methods. Uchida and McPhee [140, 141] formulated the EOM of a vehicle symbolically using Maple. They used a topological formulation combined with symbolic coordinate reduction through embedding, and the constraint equations were symbolically transformed into a block-triangular form. Similarly, Carpinelli et al. [19] obtained the EOM of elastokinematic MBS-suspension models symbolically with Maple, including the Jacobians required for the utilized linear-implicit integration scheme. Furthermore, kinematic and dynamic equations were generated with Maple by Masoudi et al. [73] and for the real-time object-oriented modeling method by Kracht et al. [66, 70]. Another symbolic generator tool for multibody systems, called Robotran, was used by Docquier et al. [37] to derive the EOM of a vehicle model with MBS-suspension as an explicit function for the generalized accelerations. The multibody model was coupled with a hydraulic model of a semi-active suspension system. Furthermore, the software

packages IPG CarMaker [122] and Dymola [63] use symbolic computation methods to obtain efficient simulation code for real-time capable vehicle models with MBS-suspension. Symbolic computation was also used in [44, 49] to perform velocity transformations.

Efficient matrix operations and matrix libraries Linear algebra operations, in particular large matrix multiplications and solution of linear and nonlinear equation systems, often consume a considerable amount of computation time in vehicle dynamics simulation [108, 127]. In the analyzed publications, we found several implementations of efficient solution algorithms based on the numerical properties of matrices and their symbolic pattern that arises from the vehicle system's topology. Furthermore, sparse matrix techniques, which exploit the sparsity of matrices and only consider their non-zero entries, were applied. Linear algebra operations were also accelerated through the use of matrix libraries with efficient algorithms and implementations for both dense and sparse matrices.

Choi et al. [22] exploited the topology of a vehicle model with kinematic MBS-suspension using a special relative joint coordinate numbering scheme to reorder the coefficient matrix of the index-1 DAE into a doubly bordered block diagonal matrix form. The DAE could then be solved either with a sparse Cholesky decomposition or a recursive block mass matrix method, reducing computation times by up to 14%. The authors of the present paper presented a block matrix approach to exploit the topology of elastokinematic MBS-suspensions within a global EOM formulation [2]. They grouped the suspension bodies into different subsystems and obtained a coefficient matrix in block-tridiagonal form, in which each block was related to a subsystem or the connection between two subsystems. The linear equation system was decomposed into several smaller systems, which were solved sequentially, reducing computation time by up to 14%. Pan et al. [87] presented an efficient solution strategy for linear equation systems arising in the explicit 4th order Runge-Kutta integration method. As the mass matrix changes only slightly between certain integrator stages, they reused its decomposition in an iterative refinement solution strategy. The computational time for a passenger vehicle was reduced by about 2.5% and for a semitrailer truck by about 9%.

Sparse matrix technology is particularly attractive for global EOM formulations, in which a high number of coordinates and potentially constraint equations results in large equation systems with sparse coefficient matrices. Cuadrado et al. [29] took advantage of sparse matrix technology to solve all linear equation systems in global index-3 classical and augmented Lagrangian formulations for vehicles with kinematic MBS-suspension, see also [26, 30–32, 40, 78–80]. Hidalgo and García de Jalón [49] employed sparse matrix techniques for truck models with suspensions containing bushings. Significant efficiency improvements were achieved in the numerical computation of the stiffness and damping matrix, and in the computation and LU-decomposition of the coefficient matrix required for HHT integration. Computation times were reduced by up to 99%, making sparse matrix technique essential to achieve real-time capability in this case.

The effect of different matrix libraries in global formulations of vehicle models with kinematic MBS-suspension was evaluated by Shiiba et al. [127, 128, 135]. Both sparse matrix and LAPACK libraries helped to reduce computation times, with a slight advantage for the Lapack libraries. However, the effect was limited for small-size vehicle systems. Sub-routines based on the BLAS matrix library and the sparse matrix library MA48 were used in the double-step semirecursive formulation, which involves several matrix multiplications, LU-factorization of a Jacobian and solution of the corresponding linear equation system [44, 49]. Computation time savings of up to 70% were achieved for different truck models with kinematic MBS-suspension, with a slight advantage for the sparse matrix library.

BLAS routines, implemented in the Intel oneAPI Library, were also used by the authors of the present paper to accelerate linear algebra operations in a global formulation of elastokinematic MBS-suspensions [2]. Pan et al. [33, 87, 91–94] employed Intel's Math Kernel Library in the double-step semirecursive formulation for the state vector derivative calculation and numerical integration. Another matrix library, Eigen, was used for linear algebra operations in the context of a semirecursive formulation [114].

Parallel processing Computational advances, especially those related to large memory and parallelization resources, have motivated the development of parallelization techniques to speed up multibody simulations [49]. A detailed discussion of the benefits and challenges of parallelization can be found in [49]. Vehicle dynamics problems are typically of a moderate size, have a high number of short kinematic closed-loops and require intensive cyclic computations. Applying parallelization techniques is not straightforward because of the computational overhead associated with creating, destroying and synchronizing threads. It is crucial to address bottlenecks that account for a significant portion of the total computation times to gain substantial benefits from parallelization.

Cuadrado et al. [29] studied the advantages of automatic parallelization in global and topological EOM formulations of a vehicle model with kinematic MBS-suspension. The global formulations experienced significant improvements in computation times, particularly the augmented Lagrangian formulation with a reduction of almost 30%. In contrast, the topological fully-recursive formulation is more suited for sequential processing and did not benefit from parallelization. Hidalgo and García de Jalón [49] analyzed the effect of parallelization in both a topological semirecursive and a global formulation, applied for vehicle models with kinematic and elastokinematic MBS-suspension, respectively. The required Jacobians of the generalized forces were computed numerically by means of finite differences, in order to avoid complicated analytical expressions and include more complex force elements. To compensate for the high costs of numerical differentiation, the columns of the Jacobians were calculated in parallel. Depending on the formulation and size of the multibody system, the Jacobian computation was sped up by a factor of 2.5 to 3.5. Furthermore, Rodríguez et al. [114] presented hardware acceleration techniques for a semirecursive formulation, including parallel computation of the mass matrix and the dependent coordinate recovery. A more coarse-grained approach for parallelization was presented by Prescott et al. [24, 105, 108]. They subdivided a full vehicle model with elastokinematic MBS-suspension by splitting the vehicle body into two different bodies, connected through a stiff bushing called ModelLink. The front and rear part of the vehicle were then computed in parallel on different processor cores. This approach was also used in [45]. Similarly, a vehicle model with elastokinematic MBS-suspension implemented in the software package Dymola was subdivided by introducing decoupling elements. This resulted in several smaller nonlinear equation systems instead of one large system, allowing them to be solved in parallel on multi-core processors. Parallel computation was also applied in vehicle models with elastokinematic MBS-suspensions implemented in the software packages SIMPACK [8, 142] and CarMaker [122].

4.2.5 Overview of real-time elastokinematic MBS-suspension models

As analyzed in the previous sections, a wide variety of EOM formulation and integration methods were utilized for real-time simulation of models with MBS-suspensions. In addition, several model simplifications and approaches in the EOM formulation and integration as well as their implementation aimed at reducing computation times. Given the increased

research on the incorporation of elastokinematic properties in recent years (see Fig. 7), this section provides an overview of the identified MBS-suspension models with elastokinematic properties. Here, we look at the information provided in the previous sections from a different perspective, analyzing the extent to which elastokinematic properties were considered and the methods employed for real-time simulation. Table 1 summarizes all 31 publications considering elastokinematic properties, which will be described in more detail in the following. They are categorized into models with general compliance, models that focus

Table 1 Overview of MBS-suspension models with elastokinematic properties in the analyzed publications

	Vehicle model with MBS-suspension		MBS-suspension	Bushings	Flexible suspension bodies	Flexible vehicle body	General compliance (not further specified)		EOM formulation (Global, Topological)	EOM	Integration method (Explicit, Implicit, Linear-Implicit)	Integration step size in ms	Massless link approach	Force-coupling method	Superimposed elastic compliance	Simplified modeling of bushings (Stiffness/Damping - Linear/Nonlinear -	Translational/Rotational/suspension Link direction)	Simplified modeling of flexible bodies (Lumped Mass, Discrete Body Dynamics, Modal Representation)	Smooth velocity approximation	Additional spring-damper elements	Low-pass filtering	Modular formulation of the EOM	Chassis-based kinematics formulation	Approximation of constraint equations	Constant Jacobian and coefficient matrices	Symbolic computation	Efficient matrix operations and matrix libraries	Parallel processing
Salaani et al. [119]	x			x	x		n.a.	n.a.	n.a.	n.a.	x	x	LM															
Pankiewicz and Rulka [97, 118]	x	x		x			T	ODE	E	1	x	x	S-L-L															
Kim et al. [59, 61, 62]	x	x					T	DAE	E	n.a.			x	S-L-T							x							
Lefèvre et al. [70]		x	x				T	ODE	E	n.a.			SD-N-TR							x	x							
Burgermeister et al. [16]	x	x					T	DAE	I	n.a.			n.a.						x						x			
Hidalgo and García de Jalón [49]	x	x					G	DAE	I	1			n.a.												x	x		
Prescott et al. [24, 45, 105, 108, 109]	x	x		x			G	DAE	I	1;2			n.a.	LM												x		
CarMaker [48, 122]	x	x					T	ODE	n.a.	n.a.			n.a.												x	x		
Dymola [63]	x	x					n.a.	n.a.	I	2			n.a.												x	x		
Rill [111]		x	x				T	ODE	LI	2			SD-N-T															
Butz et al. [17]		x	x				T	ODE	LI	n.a.			S-L-L															
Carpinelli et al. [19]		x	x				G	ODE/DAE	LI	1			SD-L-TR												x			
Archut and Corves [2]		x	x				G	ODE	LI	1	x		SD-N-TR												x	x		
Lugrís et al. [71]	x		x				G/T	DAE	I	10			MR															
Schmitt and Seifried [123, 124]	x			x			G/T	DAE	LI	10;16	x		MR															
Lee et al. [55, 69]	x	x	(x)				G	DAE	I	1;3			S-L-T	MR										x	x	x		
Franco et al. [43]	x	x	x	x			G	ODE	E	n.a.			SD-L-L	DBD														
SIMPACK [8, 9, 77, 142]	x	x	(?)	x			T	ODE	LI/?	1;?	x/?		n.a.	MR													x/?	

on the incorporation of bushings, models that incorporate flexible bodies and models that incorporate both bushings and flexible bodies (cf. Table 1).

General compliance The earliest model among the analyzed publications with consideration of elastokinematic properties is a heavy tractor model presented by Salaani et al. [119]. The torsional compliance of the chassis was modeled with the lumped mass approach, dividing the chassis into two parts with a torsional spring in between. The compliant steer effect of the suspension was superimposed on the kinematic steering as a general compliance described by a global stiffness. The suspension was further simplified, replacing its tie rods by constraints according to the massless link approach. Information about the EOM formulation and integration was not given.

Incorporation of bushings The majority of the publications summarized in Table 1 focused on the incorporation of bushings only. It should be noted that not all of them provided details of the employed bushing models so that their complexity remains unknown.

Several authors proposed and utilized model simplifications to reduce the size, complexity and numerical stiffness of the MBS-suspension models for achieving real-time capability. Pankiewicz and Rulka [97, 118] presented the macro-joint approach, which neglects the mass and inertia of all suspension links and describes the guidance of the wheels relative to the vehicle body solely through constraints, according to the massless link approach. These so-called macro-joints were expressed in independent joint coordinates using coordinate partitioning, allowing the EOM to be formulated as ODE. Elastokinematic effects were considered to a limited extent, either as quasistatic deformations of bushings between the suspension links and the vehicle body in the direction of those links, or by superimposing an overall suspension compliance on the kinematic wheel motion. These simplifications reduce the model size and avoid high-frequency oscillations, allowing stable real-time integration with the explicit Euler method. The low-frequency motion of the suspension is well represented, but the higher-frequency dynamics is not captured by the macro-joint approach.

Another quasistatic approach to account for bushing effects was proposed by Kim et al. [59, 61, 62]. They extended a kinematic MBS-suspension model, employed within the subsystem synthesis method for modular formulation of the EOM, with a superimposed compliance. The differential-algebraic EOM of the kinematic MBS-suspension were formulated topologically and integrated with the explicit Adams-Bashforth method. In each integration step, compliant bushing deformations were superimposed by shifting the hard points of the associated joints. The displacements were calculated based on the linear stiffness properties of the bushings and the joint forces. This approach improved the accuracy compared to a purely kinematic MBS-suspension, but with remaining deviations even in the quasistatic wheel travel motion.

An MBS-suspension model with bushings based on the real-time object-oriented modeling method by Kracht [65] was presented in [70]. This method treats the vehicle system as a composition of individual elements, modeled as objects and assembled modularly. The underlying equations are obtained and simplified symbolically. In the presented model, all suspension bodies were connected through rather complex bushing force elements with nonlinear stiffness- and damping properties in all directions. To reduce the numerical stiffness of the ordinary differential EOM and enable real-time integration with the explicit Bogacki-Shampine method, additional spring-damper elements and low-pass filters were introduced into the model. These additional elements required an optimization-based parametrization to maintain a good fit to physical suspension test rig measurements within a frequency range up to 15 Hz. The real-time object-oriented modeling method was also extended to consider

flexible suspension bodies with simple geometry, coarse FE meshing and simplified elastodynamics modeling [65]. However, this publication was not included in the present PRISMA analysis as it was not part of the records found in the consulted databases (see Sect. 3).

A more mathematical way to reduce the numerical stiffness of a suspension system with bushings was presented by Burgermeister et al. [16]. They neglected the inertia effects of fast-moving bodies using smooth velocity approximation, permitting larger step sizes of the employed backward differentiation formula and reducing computation times by up to 50%. However, this method may decrease the accuracy of the model as it neglects the suspension rods' inertia.

In the previous examples, the models were modified or simplified to achieve real-time capability. In contrast, other approaches focus on the EOM formulation and integration as well as the computational implementation. Hidalgo and García de Jalón [49] utilized matrix libraries and parallel processing to reduce computation times. They applied a global EOM formulation to introduce bushings in a general way and used the implicit HHT-method to address the high numerical stiffness. Furthermore, Jacobians of the generalized forces were computed numerically to avoid complicated analytical expressions and include more complex force elements. The sparsity of the matrices in the global formulation was exploited by sparse matrix techniques, and the columns of the Jacobians were computed in parallel. Computation times were reduced by about 98%, making sparse matrix techniques and parallel computation the keys to real-time capability for the considered vehicle models.

Prescott et al. [24, 105, 108] presented a combination of different approaches in the EOM formulation and integration as well as their implementation to achieve real-time capability. They used a global formulation and an implicit integration method, but unlike Hidalgo and García de Jalón [49], the Jacobians were calculated analytically for reduced computational effort. Parallel processing was realized with a more coarse-grained approach, cutting the vehicle body and dividing the vehicle into two parts computed in parallel. The coupling was maintained through a force element between the front and rear part of the vehicle body, making this approach also suitable to replicate a flexible vehicle body with a lumped mass model. The index-1 differential-algebraic EOM were integrated with the implicit backward differentiation formula. To ensure deterministic computational effort and meet hard real-time requirements with a step size of 1 ms, the number of iterations per integration step was limited. These methods were implemented in the software package LMS Virtual Lab Motion, and were also applied by Ramirez Ruiz et al. [109] and Grottooli et al. [45]. The methods were also extended for real-time simulations with flexible bodies modeled with the component mode synthesis technique based on FE models [106, 107]. To achieve real-time capability, a co-simulation approach with separate integration of the rigid and the flexible portion of the equations was proposed. However, details of the suspension models and simulation results were not given in the publications. These publications were not part of the present PRISMA analysis, as they are not full papers or not indexed in the consulted databases.

Real-time simulation of vehicles with MBS-suspension containing bushings in the commercial special-purpose software CarMaker was presented in [48, 122]. The EOM were generated symbolically in minimal coordinates and simplified to generate compact equations. Additionally, parallel processing was utilized. Further details about the EOM formulation, integration, or implementation were not given.

A vehicle model with suspension bushings, implemented in the software package Dy-mola, was presented by Klomp et al. [63]. Real-time capability was achieved through inline integration, i.e., the discretization formulas of the implicit Euler scheme were merged with the model equations and symbolically optimized for computational efficiency. Furthermore,

the computation was parallelized on the coarse-grained level by decoupling the front and rear suspension, the left and right side of the suspension linkages, and the wheels. Real-time performance was achieved with an integration step size of 2 ms.

Another approach to real-time simulation of vehicle models with elastokinematic properties are linear-implicit integration methods. Rill [111] presented a modified implicit Euler method that does not require iterative solution methods. He applied it to a simple planar suspension model, consisting of three bodies and a bushing with nonlinear stiffness and damping properties. The required Jacobians were calculated analytically in each integration step. Simulations of vehicle maneuvers with a step size of 2 ms compared well with reference solutions obtained with an accurate implicit solver.

The linear-implicit Euler method was also used by Butz et al. [17] for real-time simulation of a suspension with bushing force elements. The model was used in combination with optimization methods to estimate the stiffness coefficients of the bushings based on virtual suspension test rig measurements. The bushings were modeled in a simplified way, only taking their linear stiffness in direction of the suspension links into account.

Carpinelli et al. [19] compared the efficiency of the linear-implicit Euler method in a global DAE formulation in dependent coordinates and an ODE formulation in independent coordinates. The ODE were derived from the DAE through independent coordinate switching based on nullspace transformation. The EOM and Jacobians required for the linear-implicit Euler integration were computed symbolically, considering only the stiff terms in the vector of generalized forces. Two different suspension models were analyzed: In the *ideal joints model*, the suspension bodies were connected through ideal joints resulting in a high number of constraints; in the *bushing model*, the suspension's elastokinematics was modeled through bushings with linear stiffness and damping properties in all directions, resulting in a high number of force elements and low number of constraints. The coordinate switching from DAE to ODE formulation was beneficial for the *ideal joints model* but increased the computation times of the *bushing model*. However, the *bushing model* was faster than the *ideal joints model* with both formulations. The authors concluded that coordinate reduction only seems to be attractive for highly constrained systems.

A global formulation in natural coordinates was also employed by the authors of the present paper, who presented detailed suspension models with complex bushing force elements featuring nonlinear stiffness and damping properties in all directions [2]. Utilizing the force-coupling method, they replaced the few joints in the systems, e.g., those in the spring-damper units, with appropriately parameterized bushings. The ordinary differential EOM were integrated with a higher-order two-stage linear-implicit integration scheme. The required Jacobians of the generalized forces were obtained with an analytical online-linearization procedure. Several approaches to enhance the efficiency of the integration scheme were proposed: The suspension topology was exploited to rearrange the coefficient matrix of the involved linear equation system into a block matrix form. This allowed to decompose the equation system into smaller systems that were solved subsequently with reduced computational effort. Linear algebra operations were further accelerated using BLAS matrix library. In addition, the constant Jacobian and coefficient matrix approach was applied, updating the Jacobians at time intervals larger than the integration step size to reduce the need for expensive Jacobian calculation and matrix decomposition. Highly accurate results were obtained in real-time with an integration step size of 1 ms.

Incorporation of flexible bodies Few publications focused on the modeling of flexible bodies. Lugić et al. [71] incorporated the flexibility of suspension bodies into a full vehicle model. They employed a modal representation with mode shapes based on FE and analytical models. The modal basis was reduced through Craig-Bampton reduction to a total of

64 modes that were distributed among 12 flexible suspension bodies. Inertia shape integrals were calculated before the real-time simulation. A topological EOM formulation in relative joint coordinate proved to be almost five times faster than a global formulation in natural coordinates. The EOM were integrated with the implicit trapezoidal solver with a rather coarse integration step size of 10 ms.

Schmitt and Seifried [123, 124] presented a vehicle model with a flexible vehicle body based on a very detailed FE model. The Craig-Bampton method was used to obtain a high number of 40 shape modes, consisting of 28 inner modes and 12 boundary modes [123]. In contrast to the detailed modeling of the vehicle body, the suspension was strongly simplified using the massless link approach. All suspension links were replaced by constraints neglecting their mass and inertia, leaving only the vehicle body, the wheel mounts and the wheels as actual bodies. The EOM were formulated as DAE, either with a global approach in natural coordinates or topologically in a mixed coordinate formulation, in which the wheel motion was described relative to the wheel mount in terms of joint coordinates. The latter approach resulted in 24 coordinates and 20 constraint equations less than the global formulation, reducing computation times by 36%. The EOM were integrated as index-2 DAE with projections using the linear-implicit Euler method. The required Jacobians were calculated analytically neglecting Coriolis and centrifugal forces. Real-time simulations were performed with rather high integration step sizes of 10 ms or more.

Incorporation of bushings and flexible bodies Real-time vehicle models with wheel suspensions combining bushings and flexible suspension bodies were rarely found in the publications analyzed. Lee et al. [69] proposed a chassis-based kinematics formulation for a suspension with simple bushing elements featuring only linear stiffness properties in translational directions. The EOM, constraints equations and generalized forces were formulated with respect to the chassis-body reference frame. Thus, the system Jacobian required for the implicit integration could be approximated as constant, allowing its time-consuming calculation and decomposition to be performed in a preprocessing step before the actual real-time simulation. This reduced computation times by about 65% but was also associated with a reduction in accuracy [69]. The model was extended to integrate flexible suspension bodies in combination with bushings, implementing a coupled torsion beam axle with a flexible torsion beam using modal representation [55].

Franco et al. [43] proposed the discrete body dynamics method for modeling vehicles with flexible suspension bodies and bushings. They replaced the suspension links with a set of virtual spring-damper elements that combine the longitudinal flexibility of the suspension links and the connected bushings. The mass of the suspension links was added to the vehicle body and the wheel mounts. The EOM were formulated directly as ODE and integrated with an explicit integration method. However, this method involved significant model simplifications, resulting in only fair accuracy compared to a detailed reference simulation.

Vehicle models combining suspension bushings and a flexible vehicle body were implemented in the commercial software package SIMPACK [8, 9, 77, 142]. They were based on high-fidelity offline vehicle models and simplified for real-time capability, with the most detailed information provided in [142]. The force-coupling method was applied to remove algebraic constraints and obtain the EOM as ODE. A linear-implicit integration scheme with a step size of 1 ms was used for numerically stable integration, and further reduction of computation times was achieved through parallel processing. The flexible vehicle body was modeled as FE model and reduced by means of condensation, whereby the number of modes was decreased to achieve the real-time goal. The reduced flexible vehicle body model yielded significantly better results than a rigid one, especially in the frequency range above 20 Hz.

5 Conclusion

Vehicle dynamics simulation allows for predicting the dynamic behavior of road vehicles in order to improve their driving characteristics, ride comfort and safety. Real-time applications have become standard tools in the automobile industry, in particular driving simulators playing a key role in vehicle dynamics development. There is a demand for more accurate yet real-time capable vehicle dynamics models to extend the applicability and reliability of such real-time applications. This applies in particular to the detailed representation of the kinematics and elastokinematics of wheel suspensions. However, real-time requirements present challenges for detailed vehicle models due to large model sizes, closed-loop topologies and high numerical stiffness.

In the present paper, numerous scientific publications on real-time multibody simulation of road vehicles have been analyzed using the PRISMA methodology. This systematic mapping provides an overview of real-time vehicle dynamics models and their underlying methods, with a particular focus on wheel suspension modeling. A total of 765 records published between 1999 and 2024 was found through search queries in the databases Scopus, Web of Science and Google Scholar. Eligible reports were selected based on specific pre-defined criteria, resulting in a set of 109 scientific publications analyzed in detail. It should be noted that this systematic mapping is limited to scientific publications indexed in the consulted databases. Methods and implementations of commercial multibody software packages are only partially covered, as they are usually not fully published and only accessible to a limited extent.

Both, vehicle models with simple LM- and LT-suspensions and vehicle models with detailed MBS-suspensions were identified in the analyzed publications. The former were often utilized for practical applications, as they are inherently real-time capable due to their low complexity. However, they cover the kinematic and elastokinematic properties of suspensions only to a limited degree of accuracy. The majority of the publications considered MBS-suspensions, of which only one third included elastokinematic properties. These elastokinematic MBS-suspensions mainly featured the local compliance due to bushings, while the structural compliance of flexible bodies was only rarely considered. As real-time simulation with detailed MBS-suspension models is challenging, these publications focused on the research of real-time multibody simulation methods.

The EOM of the MBS-suspensions models were formulated using both global and topological formulations, depending on the characteristics of the specific suspension model. In most cases, the EOM were obtained topologically employing relative joint coordinates, usually through semirecursive formulation with velocity transformation. Global formulations were used more frequently for elastokinematic than for kinematic MBS-suspensions. This is likely because bushing force elements can be incorporated in a more general way with global formulations. Furthermore, coordinate reduction by means of coordinate partitioning was applied to remove constraint equations and express the EOM as ODE, both in global and in topological formulations.

The numerical methods for integrating the EOM of the MBS-suspensions in real-time vary, with explicit and implicit methods almost equally distributed. Explicit methods were mainly employed for kinematic MBS-suspensions. In some cases, they were also used for elastokinematic MBS-suspensions, requiring model simplifications to avoid high numerical stiffness. Implicit methods are suited for both kinematic and elastokinematic MBS-suspensions. They were typically employed with step sizes in the range of ten of ms and in the range of ms, respectively. To make their computational effort deterministic and improve their suitability for real-time simulations, the number of iterations per integration step

was limited in some approaches. Linear-implicit integration methods were rarely used. They have proven to be well suited for real-time simulation of elastokinematic MBS-suspensions, because they combine high numerical stability with constant computational effort per integration step, without the need for iterative solution methods.

Various simplifications in the modeling of MBS-suspensions were found in the analyzed publications. Techniques such as the massless link approach, the rod-removal technique and the force-coupling method aim to reduce the model size in terms of number of coordinates and constraint equations, as well as to remove closed kinematic loops. Other model simplifications employed focus on reducing computational complexity and numerical stiffness, especially when modeling elastokinematic properties. While model simplifications help to reduce computation times, they are often associated with a decrease in model accuracy.

Furthermore, several approaches to reduce computation times were identified in the EOM formulation and integration. They mainly aim to avoid or reduce the need for expensive Jacobians computations, be it the Jacobians of the constraint equations or the generalized forces, and for the expensive decomposition of coefficient matrices of linear equation systems. In the implementation, efficient linear algebra operations, matrix libraries and sparse matrix techniques were utilized to accelerate linear algebra calculations. Symbolic computation methods were used to create simple and compact equations, and parallel processing techniques were implemented to compute specific calculations but also entire vehicle subsystems in parallel.

The elastokinematics of suspensions strongly affects the driving characteristics of vehicles and is specifically designed for good driving comfort and handling. Incorporating elastokinematic properties into detailed MBS-suspension models is challenging, as it is associated with increased model complexity and numerical stiffness.

Most of the elastokinematic MBS-suspensions only considered bushing elements, which in many cases were even simplified for reduced complexity. The most complex bushing elements employed have nonlinear stiffness and damping properties in translational and rotational directions. Frequency-dependent properties were not considered. In addition, many publications did not provide detailed information about the employed bushing elements.

Some modeling approaches for elastokinematic MBS-suspension models considered bushing effects only quasistatically and superimposed them on the kinematic wheel motion. Others aimed at reducing the numerical stiffness through neglecting the mass and inertia of the suspension links or introducing additional model elements like spring-damper elements or low-pass filters. There are also some approaches for real-time simulation of MBS-suspensions with bushing elements without major model simplifications, mainly utilizing linear-implicit integration schemes, efficient linear algebra implementations or parallel processing.

Elastic deformations of flexible bodies were considered only in very few vehicle models with MBS-suspensions. Typical modeling approaches are lumped mass models or modal representation based on analytical or FE models. The flexibility of the vehicle body was considered more often than those of the suspension bodies, in particular in combination with bushing elements.

Based on these findings on real-time capable elastokinematic MBS-suspension models, the inclusion of more complex bushings elements, e.g., with frequency-dependent properties, and the detailed consideration of body flexibilities, especially in combination with bushings, are interesting for the future. This will help to increase the applicability and reliability of real-time vehicle dynamics models for extended handling and comfort analyses.

Author contributions J.-L.A. and B.C. had the idea for the article. J.-L.A. performed the literature search and data analysis. J.-L.A. drafted the manuscript. B.C. and J.-L.A. critically revised the manuscript. All authors approved the version of the manuscript to be published.

Funding information Open Access funding enabled and organized by Projekt DEAL.

Data availability Data will be made available on request.

Declarations

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

1. Adamski, D.: Simulation in der Fahrwerktechnik: Einführung in die Erstellung von Komponenten- und Gesamtfahrzeugmodellen. ATZ / MTZ-Fachbuch. Springer, Wiesbaden (2014). <https://doi.org/10.1007/978-3-658-06536-2>
2. Archut, J.L., Corves, B.: Real-time multibody simulation of vehicle wheel suspensions of different topologies with elastokinematic properties. Multibody Syst. Dyn. (2024). <https://doi.org/10.1007/s11044-024-09984-2>
3. Arnold, M., Burgermeister, B., Eichberger, A.: Linearly implicit time integration methods in real-time applications: DAEs and stiff ODEs. Multibody Syst. Dyn. **17**, 99–117 (2007). <https://doi.org/10.1007/s11044-007-9036-8>
4. Arnold, M., Burgermeister, B., Führer, C., Hippmann, G., Rill, G.: Numerical methods in vehicle system dynamics: state of the art and current developments. Veh. Syst. Dyn. **49**(7), 1159–1207 (2011). <https://doi.org/10.1080/00423114.2011.582953>
5. Asthana, S.V., Rasal, S., Vellandi, V., Uil, R., Lanas, G., Tosolin, G.: Virtual Tire Development for New Electric Vehicle through Driver in Loop Approach. SAE Technical Paper 2024-01-2654 (2024) <https://doi.org/10.4271/2024-01-2654>
6. Bae, D.S., Lee, J.K., Cho, H.J., Yae, H.: An explicit integration method for realtime simulation of multibody vehicle models. Comput. Methods Appl. Mech. Eng. **187**(1–2), 337–350 (2000). [https://doi.org/10.1016/S0045-7825\(99\)00138-3](https://doi.org/10.1016/S0045-7825(99)00138-3)
7. Barbosa, F., Bitencourt, R., Costa, G., Tomich, J., Couto, A., Silva, R., Horta, J.: Virtual model correlation and its validation using a dynamic driver simulator. SAE Technical Paper Series 2020-36-0179 (2021) <https://doi.org/10.4271/2020-36-0179>
8. Bauer, F., Fleischhacker, J.: Hardware-in-the-Loop Simulation of Electro-Pneumatic Brake Systems. SAE Technical Paper 2015-01-2745 (2015) <https://doi.org/10.4271/2015-01-2745>
9. Bauer, F., Lion, A., Litter, S., Topčagić, E.: Hardware-in-the-loop system for electro-pneumatic brake systems of commercial vehicles. In: Rosenberger, M., Plöchl, M., Six, K., Edelmann, J. (eds.) The Dynamics of Vehicles on Roads and Tracks, pp. 358–366. Chapman and Hall/CRC, Boca Raton (2016). <https://doi.org/10.1201/b21185-38>
10. Baumgarte, J.: Stabilization of constraints and integrals of motion in dynamical systems. Comput. Methods Appl. Mech. Eng. **1**(1), 1–16 (1972). [https://doi.org/10.1016/0045-7825\(72\)90018-7](https://doi.org/10.1016/0045-7825(72)90018-7)
11. Bayo, E., Ledesma, R.: Augmented Lagrangian and mass-orthogonal projection methods for constrained multibody dynamics. Nonlinear Dyn. **9**(1–2), 113–130 (1996). <https://doi.org/10.1007/BF01833296>
12. Blundell, M., Harty, D.: The Multibody Systems Approach to Vehicle Dynamics, 2nd edn. Elsevier, Amsterdam (2015). <https://doi.org/10.1016/C2012-0-07000-7>
13. Bruni, S., Meijaard, J.P., Rill, G., Schwab, A.L.: State-of-the-art and challenges of railway and road vehicle dynamics with multibody dynamics approaches. Multibody Syst. Dyn. **49**(1), 1–32 (2020). <https://doi.org/10.1007/s11044-020-09735-z>

14. Buchta, R., Liu-Henke, X.B.: Modeling of a Battery Electric Vehicle (BEV) for efficient model-based design. *Solid State Phenom.* **198**, 533–538 (2013). <https://doi.org/10.4028/www.scientific.net/SSP.198.533>
15. Burgermeister, B., Arnold, M., Esterl, B.: DAE time integration for real-time applications in multi-body dynamics. *Z. Angew. Math. Mech.* **86**(10), 759–771 (2006). <https://doi.org/10.1002/zamm.200610284>
16. Burgermeister, B., Arnold, M., Eichberger, A.: Smooth velocity approximation for constrained systems in real-time simulation. *Multibody Syst. Dyn.* **26**(1), 1–14 (2011). <https://doi.org/10.1007/s11044-011-9243-1>
17. Butz, T., Simeon, B., Stadler, M.: Optimal design of experiments for estimating parameters of a vehicle dynamics simulation model. *J. Comput. Nonlinear Dyn.* **5**(3), 031011 (2010). <https://doi.org/10.1115/1.4001391>
18. Candreva, S., Mundo, D., Gubitosa, M., Toso, A.: On the correlation between a 3D high-fidelity multi-body vehicle model and a 1D 15-DOFs vehicle concept model. In: ISMA International Conference on Noise and Vibration Engineering. Leuven, Belgium (2014)
19. Carpinelli, M., Gubitosa, M., Mundo, D., Desmet, W.: Automated independent coordinates' switching for the solution of stiff DAEs with the linearly implicit Euler method. *Multibody Syst. Dyn.* **36**(1), 67–85 (2016). <https://doi.org/10.1007/s11044-015-9455-x>
20. Choi, G.J., Yoo, Y.M., Lee, K.P., Yoon, Y.S.: Vehicle modeling methods for real-time dynamic simulation using suspension composite joints *. *Mech. Struct. Mach.* **28**(4), 303–321 (2000). <https://doi.org/10.1081/SME-100102024>
21. Choi, G.J., Yoo, Y.M., Lees, K.P., Yoon, Y.S.: A real-time multibody vehicle dynamic analysis method using suspension composite joints. *Int. J. Veh. Des.* **24**(2/3), 259 (2000). <https://doi.org/10.1504/IJVD.2000.005184>
22. Choi, G., Yoo, Y., Im, J.: Efficient solving methods exploiting sparsity of matrix in real-time multi-body dynamic simulation with relative coordinate formulation. *KSME Int. J.* **15**(8), 1090–1096 (2001). <https://doi.org/10.1007/BF03185089>
23. Cianetti, F., Garzia, R., Palmieri, M., Ambrogi, F., Braccresi, C.: An estimation model of suspension loads in explicit multibody simulation. *IOP Conf. Ser., Mater. Sci. Eng.* **1038**(1), 012042 (2021). <https://doi.org/10.1088/1757-899X/1038/1/012042>
24. Cibrario, V., Bäcker, M., De Cuyper, J., Donders, S., Toso, A., Prescott, W., Gallrein, A., Gizatullin, A.: Real-time technologies in driving dynamics performance simulation through full vehicle multibody, 1D chassis subsystems and tire modeling integration. In: World Automotive Congress 2014. Maastricht, The Netherlands (2014)
25. Cipelli, M., Schiehlen, W., Cheli, F.: Driver-in-the-loop simulations with parametric car models. *Veh. Syst. Dyn.* **46**, 33–48 (2008). <https://doi.org/10.1080/00423110701882280>
26. Cuadrado, J., Dopico, D.: A hybrid global-topological real-time formulation for multibody systems. In: Proceedings of the 2003 ASME Design Engineering Technical Conferences and Computers and Information in Engineering Conference, pp. 115–121. ASME, Chicago (2003). <https://doi.org/10.1115/DETC2003/VIB-48315>
27. Cuadrado, J., Dopico, D.: A combined penalty and semi-recursive formulation for closed-loops in MBS. In: Huang, T. (ed.) Proceedings of Eleventh World Congress in Mechanism and Machine Science. China Machine Press, Tianjin (2004)
28. Cuadrado, J., Cardenal, J., Bayo, E.: Modeling and solution methods for efficient real-time simulation of multibody dynamics. *Multibody Syst. Dyn.* **1**(3), 259–280 (1997). <https://doi.org/10.1023/A:1009754006096>
29. Cuadrado, J., Cardenal, J., Morer, P., Bayo, E.: Intelligent simulation of multibody dynamics: space-state and descriptor methods in sequential and parallel computing environments. *Multibody Syst. Dyn.* **4**(1), 55–73 (2000). <https://doi.org/10.1023/A:1009824327480>
30. Cuadrado, J., Dopico, D., Gonzalez, M., Naya, M.A.: A combined penalty and recursive real-time formulation for multibody dynamics. *J. Mech. Des.* **126**(4), 602–608 (2004). <https://doi.org/10.1115/1.1758257>
31. Cuadrado, J., Dopico, D., Naya, M.A., Gonzalez, M.: Penalty, semi-recursive and hybrid methods for MBS real-time dynamics in the context of structural integrators. *Multibody Syst. Dyn.* **12**(2), 117–132 (2004). <https://doi.org/10.1023/B:MUBO.0000044421.04658.de>
32. Cuadrado, J., Naya, M.A., Dopico, D., Gonzalez, M.: Application of Real-Time Multibody Dynamics to Vehicle Control. *Bull. Transilv. Univ. Braşov* (2006)
33. Dai, W., Pan, Y., Min, C., Zhang, S.P., Zhao, J.: Real-time modeling of vehicle's longitudinal-vertical dynamics in ADAS applications. *Actuators* **11**(12), 378 (2022). <https://doi.org/10.3390/act11120378>
34. Dandekar, R., Kracht, F.E.: Generalized method for real-time object-oriented modeling and simulation of systems applied to a vehicle wheel suspension mechanism. In: Badodkar, D.N., Dwarakanath, T.A.

- (eds.) *Machines, Mechanism and Robotics. Lecture Notes in Mechanical Engineering*, pp. 463–474. Springer, Singapore (2019). https://doi.org/10.1007/978-981-10-8597-0_39
35. Dessort, R., Meissner, M., Kolmeder, S., Prokop, G., Kubenz, J.: Experimental nonlinear system identification of a shock absorber focusing on secondary ride comfort. In: Pfeffer, P. (ed.) *12th International Munich Chassis Symposium 2021, ATZ Live*, pp. 218–240. Springer, Berlin (2022). https://doi.org/10.1007/978-3-662-64550-5_13
 36. Dias, C.A.R., Campolina, M.A.M., Naves, E.E.P., Júnior, J.L.: Steering ratio analysis in a vehicle dynamics simulator using a kinematics and compliance correlated mode. *SAE Technical Paper Series 2020-36-0232* (2021) <https://doi.org/10.4271/2020-36-0232>
 37. Docquier, N., Poncelet, A., Delannoy, M., Fiset, P.: Multiphysics modelling of multibody systems: application to car semi-active suspensions. *Veh. Syst. Dyn.* **48**(12), 1439–1460 (2010). <https://doi.org/10.1080/00423110903509335>
 38. Docquier, N., Poncelet, A., Fiset, P.: ROBOTRAN: a powerful symbolic generator of multibody models. *Mech. Sci.* **4**(1), 199–219 (2013). <https://doi.org/10.5194/ms-4-199-2013>
 39. Dong, X., Yu, M., Liao, C., Chen, W., Li, Z.: Rapid control prototyping development of intelligent control system of vehicle semi-active suspension. In: *7th World Congress on Intelligent Control and Automation*. Chongqing, China, pp. 1971–1975 (2008). <https://doi.org/10.1109/WCICA.2008.4593227>
 40. Dopico, D., Lugi, U., Gonzalez, M., Cuadrado, J.: Two implementations of IRK integrators for real-time multibody dynamics. *Int. J. Numer. Methods Eng.* **65**(12), 2091–2111 (2006). <https://doi.org/10.1002/nme.1544>
 41. Ersoy, M., Gies, S. (eds.): *Fahrwerkshandbuch: Grundlagen - Fahrdynamik - Fahrverhalten - Komponenten - elektronische Systeme - Fahrerassistenz - autonomes Fahren - Perspektiven*, 5. Auflage. ATZ/MTZ-Fachbuch. Springer, Wiesbaden (2017). <https://doi.org/10.1007/978-3-658-15468-4>
 42. Falchi, L., Goy, F., Ludwig, C., Salgarello, A.: Feasibility study for ride comfort application by functional vehicle models: new excitation method and model extension. In: Pfeffer, P. (ed.) *12th International Munich Chassis Symposium 2021, ATZ Live*, pp. 261–274. Springer, Berlin (2022). https://doi.org/10.1007/978-3-662-64550-5_15
 43. Franco, Y., Shani, M., Gat, G., Shmulevich, I.: Three-dimensional dynamic model for off-road vehicles using discrete body dynamics. *J. Terramech.* **91**, 297–307 (2020). <https://doi.org/10.1016/j.jterra.2020.08.003>
 44. García de Jalón, J., Callejo, A., Hidalgo, A.F.: Efficient solution of Maggi's equations. *J. Comput. Nonlinear Dyn.* **7**(2), 021003 (2012). <https://doi.org/10.1115/1.4005238>
 45. Grotoli, M., van der Heide, A., Lemmens, Y.: A high fidelity driving simulation platform for the development and validation of advanced driver assistance systems. In: Yan, X.T., Bradley, D., Russell, D., Moore, P. (eds.) *Reinventing Mechatronics*, pp. 93–109. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-29131-0_7
 46. Haug, E.J.: *Computer Aided Kinematics and Dynamics of Mechanical Systems. - 1: Basic Methods*. Allyn and Bacon Series in Engineering. Allyn and Bacon, Boston (1989)
 47. He, L., Pan, Y., He, Y., Li, Z., Królczyk, G., Du, H.: Control strategy for vibration suppression of a vehicle multibody system on a bumpy road. *Mech. Mach. Theory* **174**, 104891 (2022). <https://doi.org/10.1016/j.mechmachtheory.2022.104891>
 48. Henning, J., Schmidt, S., Bodenstein, J.: Efficient optimisation of handling characteristics using real-time capable multi-body suspensions. *ATZ Worldw.* **117**(4), 22–25 (2015). <https://doi.org/10.1007/s38311-015-0006-0>
 49. Hidalgo, A.F., García de Jalón, J.: Real-time dynamic simulations of large road vehicles using dense, sparse, and parallelization techniques. *J. Comput. Nonlinear Dyn.* **10**(3), 031005 (2015). <https://doi.org/10.1115/1.4028794>
 50. Hussain, Z., Azlan, N.Z.: KANE's method for dynamic modeling. In: Staff, I. (ed.) *2016 IEEE International Conference on Automatic Control and Intelligent Systems (I2CACIS)*, pp. 174–179. IEEE, Selangor (2016). <https://doi.org/10.1109/I2CACIS.2016.7885310>
 51. ISO/IEC 2382: Information technology — Vocabulary (2015)
 52. Jafarzadehpour, N., Bidlingmaier, M., Corves, B.: Integrating Body-in-White Influences on Vehicle Dynamics into Real-Time Models. *SAE Technical Paper Series 2021-01-5085* (2021) <https://doi.org/10.4271/2021-01-5085>
 53. Jerkovsky, W.: The structure of multibody dynamics equations. *J. Guid. Control* **1**(3), 173–182 (1978). <https://doi.org/10.2514/3.55761>
 54. Kang, H.C., Kim, S.S., Lee, C.H.: Parallel processing with the subsystem synthesis method for efficient vehicle analysis. *J. Mech. Sci. Technol.* **29**(7), 2663–2669 (2015). <https://doi.org/10.1007/s12206-015-0512-4>
 55. Kang, J.S., Lee, J.K., Bae, D.: A chassis-based kinematic formulation for flexible multi-body vehicle dynamics. *Mech. Based Des. Struct. Mach.* **45**(1), 12–24 (2017). <https://doi.org/10.1080/15397734.2015.1126693>

56. Kersten, T., Ungemach, G., Schick, B., Böhle, M., Martynkewicz, M., Harendza, N.: Study to assess the controllability after chassis component damages on the dynamic driving simulator. In: Pfeffer, P. (ed.) 12th International Munich Chassis Symposium 2021, ATZ Live, pp. 11–42. Springer, Berlin (2022). https://doi.org/10.1007/978-3-662-64550-5_2
57. Kim, S.S.: A subsystem synthesis method for efficient vehicle multibody dynamics. *Multibody Syst. Dyn.* **7**(2), 189–207 (2002). <https://doi.org/10.1023/A:1014457111573>
58. Kim, S.S., Jeong, W.: Subsystem synthesis method with approximate function approach for a real-time multibody vehicle model. *Multibody Syst. Dyn.* **17**(2–3), 141–156 (2007). <https://doi.org/10.1007/s11044-007-9038-6>
59. Kim, S.S., Jeong, W.H.: Real-time multibody vehicle model with bush compliance effect using quasi-static analysis for HILS. *Multibody Syst. Dyn.* **22**(4), 367–382 (2009). <https://doi.org/10.1007/s11044-009-9160-8>
60. Kim, S.S., Wang, J.H.: Subsystem synthesis methods with independent coordinates for real-time multibody dynamics. *J. Mech. Sci. Technol.* **19**(S1), 312–319 (2005). <https://doi.org/10.1007/BF02916149>
61. Kim, S.S., Jeong, W.H., Kim, S.: Compliance effect consideration for real-time multibody vehicle dynamics using quasi-static analysis. In: Proceedings of the ASME International Design Engineering Technical Conferences and Computers and Information in Engineering Conference 2007. ASME, Las Vegas (2008). <https://doi.org/10.1115/DETC2007-34739>
62. Kim, S.S., Jeong, W.H., Lee, S.J., Won, M.C.: Performance evaluation of intelligent chassis controller using ecu-in-the-loop-simulator based on real-time multibody vehicle dynamics. In: The Proceedings of the 5th Asian Conference on Multibody Dynamics. Kyoto, Japan (2010). https://doi.org/10.1299/jsmeacmd.2010.5._59304-1_
63. Klomp, M., Sundström, P., Johnsson, A.: Real-time simulation of elasto-kinematic multi-body vehicle models. In: Edelmann, J., Plöchl, M., Pfeffer, P.E. (eds.) Advanced Vehicle Control AVEC'16, pp. 255–260. Crc Press, Boca Raton (2017). <https://doi.org/10.1201/9781315265285-41>
64. Kopetz, H.: Real-Time Systems: Design Principles for Distributed Embedded Applications. Real-Time Systems Series. Springer US and Imprint. Springer, New York (2011). <https://doi.org/10.1007/978-1-4419-8237-7>
65. Kracht, F.E.: Modellbildung und Simulation der Dynamik und Elastokinematik von Radaufhängungen für Echtzeitanwendungen. PhD Thesis, DuEPublico: Duisburg-Essen Publications online, University of Duisburg-Essen, Germany (2020). <https://doi.org/10.17185/DUEPUBLICO/71445>
66. Kracht, F.E., Schramm, D.: Real-time capable calculation of reaction forces of multibody systems using optimized bushings on the example of a vehicle wheel suspension. In: Kecskeméthy, A. (ed.) Multibody Dynamics 2019. Computational Methods in Applied Sciences Ser, vol. 53, pp. 409–416. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-23132-3_49
67. Lee, J.Y., Lee, W.S.: A symbolic computation method for automatic generation of a full vehicle model simulation code for a driving simulator. *J. Mech. Sci. Technol.* **19**(S1), 395–402 (2005). <https://doi.org/10.1007/BF02916160>
68. Lee, M.C., Park, M.K., Yoo, W.S., Son, K., Han, M.C.: Sliding mode control of real-time PNU vehicle driving simulator and its performance evaluation. *JSME Int. J. Ser. C* **46**(2), 557–564 (2003). <https://doi.org/10.1299/jsmec.46.557>
69. Lee, J.K., Kang, J.S., Bae, D.S.: An efficient real-time vehicle simulation method using a chassis-based kinematic formulation. *Proc. Inst. Mech. Eng. Part D, J. Automob. Eng.* **228**(3), 272–284 (2014). <https://doi.org/10.1177/0954407013507912>
70. Lefèvre, S., Kracht, F.E., Schramm, D.: Quasi-static and dynamic suspension measurements vs. multibody and real-time simulation results. In: Pfeffer, P.E. (ed.) 10th International Munich Chassis Symposium 2019. Proceedings, pp. 119–134. Springer, Wiesbaden (2020). https://doi.org/10.1007/978-3-658-26435-2_13
71. Lugiř, U., Naya, M.A., Luaces, A., Cuadrado, J.: Efficient calculation of the inertia terms in floating frame of reference formulations for flexible multibody dynamics. *Proc. Inst. Mech. Eng. Part K, J. Multi-Body Dyn.* **223**(2), 147–157 (2009). <https://doi.org/10.1243/14644193JMBD164>
72. Marciano, M., Matute, J.A., Lattarulo, R., Martí, E., Pérez, J.: Low speed longitudinal control algorithms for automated vehicles in simulation and real platforms. *Complexity* **2018**, 1–12 (2018). <https://doi.org/10.1155/2018/7615123>
73. Masoudi, R., Uchida, T., McPhee, J.: Reduction of multibody dynamic models in automotive systems using the proper orthogonal decomposition. *J. Comput. Nonlinear Dyn.* **10**(3), 031007 (2015). <https://doi.org/10.1115/1.4029390>
74. Matschinsky, W.: Radführungen der Straßenfahrzeuge Kinematik, Elasto-Kinematik und Konstruktion, 3., aktualisierte und erweiterte Auflage. VDI-Buch. Springer, Berlin (2007). <https://doi.org/10.1007/978-3-540-71197-1>

75. Mikelsons, L., Brandt, T.: Generation of continuously adjustable vehicle models using symbolic reduction methods. *Multibody Syst. Dyn.* **26**(2), 153–173 (2011). <https://doi.org/10.1007/s11044-011-9247-x>
76. Moher, D., Liberati, A., Tetzlaff, J., Altman, D.G.: Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *BMJ, Br. Med. J.* **339**, b2535 (2009). <https://doi.org/10.1136/bmj.b2535>
77. Mula, I., Tosolin, G., Akutain, X.C.: Enhanced ride comfort evaluation on the driving simulator with real-time multibody models. In: Pfeffer, P. (ed.) 12th International Munich Chassis Symposium 2021, ATZ Live, pp. 43–60. Springer, Berlin (2022). https://doi.org/10.1007/978-3-662-64550-5_3
78. Naya, M.A., Cuadrado, J.: Computer-based development of control strategies for ground vehicles. In: Rahnejat, H. (ed.) *Multi-Body Dynamics: Monitoring and Simulation Techniques III*. Professional Engineering Publishing, London (2004)
79. Naya, M.A., Cuadrado, J.: A robust tool for tuning and evaluation of automobile motion controllers. In: *ASME 2005 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, pp. 2373–2380. ASME, Long Beach (2005). <https://doi.org/10.1115/DETC2005-84668>
80. Naya, M.A., Dopico, D., Perez, J.A., Cuadrado, J.: Real-time multi-body formulation for virtual-reality-based design and evaluation of automobile controllers. *Proc. Inst. Mech. Eng. Part K, J. Multi-Body Dyn.* **221**(2), 261–276 (2007). <https://doi.org/10.1243/1464419JMBD74>
81. Nie, X., Min, C., Pan, Y., Li, Z., Królczyk, G.: An improved deep neural network model of intelligent vehicle dynamics via linear decreasing weight particle swarm and invasive weed optimization algorithms. *Sensors* **22**(13), 4676 (2022). <https://doi.org/10.3390/s22134676>
82. Nie, X., Min, C., Pan, Y., Li, K., Li, Z.: Deep-neural-network-based modelling of longitudinal-lateral dynamics to predict the vehicle states for autonomous driving. *Sensors* **22**(5), 2013 (2022). <https://doi.org/10.3390/s22052013>
83. Nikravesh, P.E., Gim, G.: Systematic construction of the equations of motion for multibody systems containing closed kinematic loops. *J. Mech. Des.* **115**(1), 143–149 (1993). <https://doi.org/10.1115/1.2919310>
84. Nybacka, M., Törlind, P., Larsson, T., Johanson, M.: Distributed real-time vehicle validation. In: *Proceedings of the ASME International Design Engineering Technical Conferences and Computers and Information in Engineering Conference - 2006*, pp. 805–812. ASME, Philadelphia (2006). <https://doi.org/10.1115/DETC2006-99154>
85. Page, M.J., McKenzie, J.E., Bossuyt, P.M., Boutron, I., Hoffmann, T.C., Mulrow, C.D., Shamseer, L., Tetzlaff, J.M., Akl, E.A., Brennan, S.E., Chou, R., Glanville, J., Grimshaw, J.M., Hróbjartsson, A., Lalu, M.M., Li, T., Loder, E.W., Mayo-Wilson, E., McDonald, S., McGuinness, L.A., Stewart, L.A., Thomas, J., Tricco, A.C., Welch, V.A., Whiting, P., Moher, D.: The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ, Br. Med. J.* **372**, n71 (2021). <https://doi.org/10.1136/bmj.n71>
86. Pagot, E., Piccinini, M., Bertolazzi, E., Biral, F.: Fast planning and tracking of complex autonomous parking maneuvers with optimal control and pseudo-neural networks. *IEEE Access* **11**, 124163–124180 (2023). <https://doi.org/10.1109/ACCESS.2023.3330431>
87. Pan, Y., García de Jalón, J.: Iterative refinement of accelerations in real-time vehicle dynamics. *J. Comput. Nonlinear Dyn.* **13**(1), 011009 (2018). <https://doi.org/10.1115/1.4037417>
88. Pan, W., Papelis, Y.E.: Real-time dynamic simulation of vehicles with electronic stability control: modelling and validation. *Int. J. Veh. Syst. Model. Test.* **1**(1/2/3), 143 (2005). <https://doi.org/10.1504/IJVSMT.2005.008577>
89. Pan, W., Papelis, Y.E., He, Y.: A vehicle-terrain system modeling and simulation approach to mobility analysis of vehicles on soft terrain. In: Gerhart, G.R., Shoemaker, C.M., Gage, D.W. (eds.) *Unmanned Ground Vehicle Technology VI*. SPIE Proceedings, pp. 520–531. SPIE, Orlando (2004). <https://doi.org/10.1117/12.547044>
90. Pan, Y., Callejo, A., Bueno, J.L., Wehage, R.A., García de Jalón, J.: Efficient and accurate modeling of rigid rods. *Multibody Syst. Dyn.* **40**(1), 23–42 (2017). <https://doi.org/10.1007/s11044-016-9520-0>
91. Pan, Y., He, Y., Mikkola, A.: Accurate real-time truck simulation via semirecursive formulation and Adams–Bashforth–Moulton algorithm. *Acta Mech. Sin.* **35**(3), 641–652 (2019). <https://doi.org/10.1007/s10409-018-0829-1>
92. Pan, Y., Xiang, S., Mikkola, A.: An efficient high-order time-step algorithm with proportional-integral control strategy for semirecursive vehicle dynamics. *IEEE Access* **7**, 40833–40842 (2019). <https://doi.org/10.1109/ACCESS.2019.2907968>
93. Pan, Y., Dai, W., Xiong, Y., Xiang, S., Mikkola, A.: Tree-topology-oriented modeling for the real-time simulation of sedan vehicle dynamics using independent coordinates and the rod-removal technique. *Mech. Mach. Theory* **143**, 103626 (2020). <https://doi.org/10.1016/j.mechmachtheory.2019.103626>

94. Pan, Y., Xiang, S., He, Y., Zhao, J., Mikkola, A.: The validation of a semi-recursive vehicle dynamics model for a real-time simulation. *Mech. Mach. Theory* **151**, 103907 (2020). <https://doi.org/10.1016/j.mechmachtheory.2020.103907>
95. Pan, Y., Nie, X., Li, Z., Gu, S.: Data-driven vehicle modeling of longitudinal dynamics based on a multibody model and deep neural networks. *Measurement* **180**, 109541 (2021). <https://doi.org/10.1016/j.measurement.2021.109541>
96. Pan, Y., Sun, Y., Li, Z., Gardoni, P.: Machine learning approaches to estimate suspension parameters for performance degradation assessment using accurate dynamic simulations. *Reliab. Eng. Syst. Saf.* **230**, 108950 (2023). <https://doi.org/10.1016/j.res.2022.108950>
97. Pankiewicz, E., Rulka, W.: From off-line to real time simulations by model reduction and modular vehicle modelling. In: *Proceedings of the 2003 ASME Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, pp. 417–424. ASME, Chicago (2003). <https://doi.org/10.1115/DETC2003/VIB-48348>
98. Parra, A., Cagigas, D., Zubizarreta, A., Rodriguez, A.J., Prieto, P.: Modelling and validation of full vehicle model based on a novel multibody formulation. In: *IECON 2019 - 45th Annual Conference of the IEEE Industrial Electronics Society*. Lisbon, Portugal, pp. 675–680 (2019). <https://doi.org/10.1109/IECON.2019.8926854>
99. Parra, A., Rodriguez, A.J., Zubizarreta, A., Perez, J.: Validation of a real-time capable multibody vehicle dynamics formulation for automotive testing frameworks based on simulation. *IEEE Access* **8**, 213253–213265 (2020). <https://doi.org/10.1109/ACCESS.2020.3040232>
100. Pastorino, R., Sanjurjo, E., Luaces, A., Naya, M.A., Desmet, W., Cuadrado, J.: Validation of a real-time multibody model for an X-by-wire vehicle prototype through field testing. *J. Comput. Nonlinear Dyn.* **10**(3), 031006 (2015). <https://doi.org/10.1115/1.4028030>
101. Peperhowe, M., Liem, K., Haupt, H.: Flexible real-time simulation of truck and trailer configurations. *SAE Int. J. Commer. Veh.* **4**(1), 232–241 (2011). <https://doi.org/10.4271/2011-01-2264>
102. Perera, H.S., Romano, R., Nunez, P.: Automated methods for converting a non real-time Cartesian multi-body vehicle dynamics model to a real-time recursive model. *SAE Technical Paper 2006-01-1165* (2006) <https://doi.org/10.4271/2006-01-1165>
103. Perrelli, M., Cosco, F., Carbone, G., Mundo, D.: Evaluation of vehicle lateral dynamic behaviour according to ISO-4138 tests by implementing a 15-DOF vehicle model and an autonomous virtual driver. *Int. J. Mech. Control* **20**(2), 31–38 (2019)
104. Piedboeuf, J.C.: Kane's equations or Jourdain's principle? In: *36th Midwest Symposium on Circuits and Systems*, pp. 1471–1474. Institute of Electrical and Electronics Engineers, Detroit (1993). <https://doi.org/10.1109/MWSCAS.1993.343389>
105. Prescott, W.: Real time simulation for high fidelity multibody dynamics and mechatronic systems. In: *Proceedings of the ASME International Mechanical Engineering Congress and Exposition - 2014*. ASME, Montreal (2014). <https://doi.org/10.1115/IMECE2014-39994>
106. Prescott, W.: Real-time simulation of flexible multibody models. In: *Proceedings of the ASME International Mechanical Engineering Congress and Exposition - 2016*. The American Society of Mechanical Engineers, Phoenix (2017). <https://doi.org/10.1115/IMECE2016-67060>
107. Prescott, W.: Simulating flexible multibody vehicle models in real time. In: *The 5th Joint International Conference on Multibody System Dynamics*. Lisbon, Portugal (2018)
108. Prescott, W., Heirman, G., Furman, M., de Cuyper, J., Dragon, L., Lippeck, A., Brauner, H.: Using High-Fidelity Multibody Vehicle Models in Real-Time Simulations. *SAE Technical Paper 2012-01-0927* (2012) <https://doi.org/10.4271/2012-01-0927>
109. Ramirez Ruiz, I., Sabbioni, E., Braghin, F., Cheli, F.: Road-vehicles control logic integrating real time multibody model follower. In: *ASME 2015 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*. Boston, Massachusetts, USA (2015). <https://doi.org/10.1115/DETC2015-47322>
110. Riegl, P., Gaull, A.: Modeling and validation of a complex vehicle dynamics model for real-time applications. In: *Proceedings of the 8th International Conference on Simulation and Modeling Methodologies, Technologies and Applications*. Porto, Portugal, pp. 403–413 (2018). <https://doi.org/10.5220/0006856304030413>
111. Rill, G.: A modified implicit Euler algorithm for solving vehicle dynamic equations. *Multibody Syst. Dyn.* **15**(1), 1–24 (2006). <https://doi.org/10.1007/s11044-006-2359-z>
112. Rill, G.: Multibody systems and simulation techniques. In: Lugner, P. (ed.) *Vehicle Dynamics of Modern Passenger Cars*. CISM International Centre for Mechanical Sciences Ser. vol. 582, pp. 309–375. Springer, Cham (2019). https://doi.org/10.1007/978-3-319-79008-4_6
113. Rill, G., Schaeffer, T., Borchsenius, F.: *Grundlagen und computergerechte Methodik der Mehrkörper-simulation*. Springer, Wiesbaden (2020). <https://doi.org/10.1007/978-3-658-28912-6>

114. Rodríguez, A.J., Pastorino, R., Carro-Lagoa, Á., Janssens, K., Naya, M.Á.: Hardware acceleration of multibody simulations for real-time embedded applications. *Multibody Syst. Dyn.* **51**(4), 455–473 (2021). <https://doi.org/10.1007/s11044-020-09738-w>
115. Romano, R.: Real-Time Multi-Body Vehicle Dynamics Using a Modular Modeling Methodology. SAE Technical Paper 2003-01-1286 (2003) <https://doi.org/10.4271/2003-01-1286>
116. Romano, R., Schultz, S.: Validation of Real-Time Multi-Body Vehicle Dynamics Models for Use in Product Design and Acquisition. SAE Technical Paper 2004-01-1582 (2004) <https://doi.org/10.4271/2004-01-1582>
117. Rong, B., Rui, X., Tao, L., Wang, G.: Theoretical modeling and numerical solution methods for flexible multibody system dynamics. *Nonlinear Dyn.* **98**(2), 1519–1553 (2019). <https://doi.org/10.1007/s11071-019-05191-3>
118. Rulka, W., Pankiewicz, E.: MBS approach to generate equations of motions for HiL-simulations in vehicle dynamics. *Multibody Syst. Dyn.* **14**(3–4), 367–386 (2005). <https://doi.org/10.1007/s11044-005-1144-8>
119. Salaani, M.K., Heydinger, G.J., Grygier, P.A.: Heavy tractor-trailer vehicle dynamics modeling for the national advanced driving simulator. *SAE Transact.* **112**, 1018–1031 (2003). <https://doi.org/10.4271/2003-01-0965>
120. Sayers, M.W.: Vehicle models for RTS applications. *Veh. Syst. Dyn.* **32**(4–5), 421–438 (1999). <https://doi.org/10.1076/vesd.32.4.421.2082>
121. Schiehlen, W., Rügauer, A., Schirle, T.: Force coupling versus differential algebraic description of constrained multibody systems. *Multibody Syst. Dyn.* **4**(4), 317–340 (2000). <https://doi.org/10.1023/A:1009864502590>
122. Schmidt, S., Schmidt, E., Schick, W., Henning, J.: Optimization of vehicle handling performance using a full vehicle model with Multi-Body System (MBS) suspensions in multiple real time - applying the DoE method. *Int. J. Automot. Eng.* **6**(2), 53–57 (2015). https://doi.org/10.20485/jsaeijae.6.2_53
123. Schmitt, A.G.: Real-time simulation of flexible multibody systems in vehicle dynamics. PhD Thesis, TUHH Universitätsbibliothek (2019). <https://doi.org/10.15480/882.2510>
124. Schmitt, A.G., Seifried, R.: Comparison of various models and integration method for real-time simulation of complex vehicle models with structural flexibility. In: Proceedings of ISMA2016, International Conference on Noise and Vibration Engineering/USD2016, International Conference on Uncertainty in Structural Dynamics. Leuven, Belgium (2016)
125. Schramm, D., Hiller, M., Bardini, R.: Modellbildung und Simulation der Dynamik von Kraftfahrzeugen, 3., aktualisierte und ergänzte Auflage. Springer, Berlin (2018). <https://doi.org/10.1007/978-3-662-54481-5>
126. Serban, R., Haug, E.J.: Globally independent coordinates for real-time vehicle simulation. *J. Mech. Des.* **122**(4), 575–582 (2000). <https://doi.org/10.1115/1.1289389>
127. Shiiba, T., Machida, N.: Efficiency evaluation of the real-time multibody analysis with matrix libraries. In: The Proceedings of the 5th Asian Conference on Multibody Dynamics. Kyoto, Japan (2010). https://doi.org/10.1299/jsmeacmd.2010.5_58291-1_
128. Shiiba, T., Murata, W.: Experimental validation of steering torque feedback simulator through vehicle running test. *J. Mech. Sci. Technol.* **23**(4), 954–959 (2009). <https://doi.org/10.1007/s12206-009-0320-9>
129. Shiiba, T., Suda, Y.: Development of driving simulator with full vehicle model of multibody dynamics. *JSAE Rev.* **23**(2), 223–230 (2002). [https://doi.org/10.1016/S0389-4304\(02\)00165-0](https://doi.org/10.1016/S0389-4304(02)00165-0)
130. Shiiba, T., Suda, Y.: Real-time multibody analysis environment for driving simulator. In: ASME 2005 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference, pp. 1791–1796. ASME, Long Beach (2005). <https://doi.org/10.1115/DETC2005-84625>
131. Shiiba, T., Suda, Y.: Evaluation of driver's behavior with multibody-based driving simulator. *Multibody Syst. Dyn.* **17**(2–3), 195–208 (2007). <https://doi.org/10.1007/s11044-007-9041-y>
132. Shiiba, T., Obana, K., Machida, N.: Comparison of linearized vs. non-linearized multibody vehicle model for real-time simulation. *Int. J. Non-Linear Mech.* **53**, 32–40 (2013). <https://doi.org/10.1016/j.ijnonlinmec.2013.01.021>
133. Solis, D., Haug, E., Shome, S.: Advanced numerical techniques for hi-fidelity on- and off-road vehicle simulation. In: Modeling and Simulation Technologies Conference and Exhibit. American Institute of Aeronautics and Astronautics, Portland (1999). <https://doi.org/10.2514/6.1999-4344>
134. Sun, Y., Pan, Y., Kawsar, I., Wang, G., Hou, L.: Combined recurrent neural networks and particle-swarm optimization for sideslip-angle estimation based on a vehicle multibody dynamics model. *Multibody Syst. Dyn.* (2024). <https://doi.org/10.1007/s11044-024-09973-5>
135. Takouda, T., Owaki, Y., Tsushima, M., Shiiba, T.: Performance improvement in steering torque simulator with multibody vehicle model. In: ASME 2013 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference. ASME, Portland (2013). <https://doi.org/10.1115/DETC2013-12131>

136. Tang, F., Wei, Y.D., Zhou, X.J., Luo, Z.H., Xie, M.X., Li, P.X.: Research on vehicle dynamics simulation for driving simulator. *Adv. Mater. Res.* **308–310**, 1946–1950 (2011). <https://doi.org/10.4028/www.scientific.net/AMR.308-310.1946>
137. Torres-Moreno, J.L., Blanco-Claraco, J.L., Giménez-Fernández, A., Sanjurjo, E., Naya, M.Á.: Online kinematic and dynamic-state estimation for constrained multibody systems based on IMUs. *Sensors* **16**(3), 333 (2016). <https://doi.org/10.3390/s16030333>
138. Tsushima, M., Kitahara, E., Shiiba, T., Motosugi, T.: Development of hardware-in-the-loop simulation system for steering evaluation using multibody kinematic analysis. *SAE Int. J. Passeng. Cars, Mech. Syst.* **7**(2), 497–504 (2014). <https://doi.org/10.4271/2014-01-0086>
139. Tsushima, M., Morihara, K., Shiiba, T.: Steering HILS system with multibody kinematics model considering suspension compliance steer. In: Rosenberger, M., Plöchl, M., Six, K., Edelmann, J. (eds.) *The Dynamics of Vehicles on Roads and Tracks*. Chapman and Hall/CRC, Boca Raton (2016)
140. Uchida, T., McPhee, J.: Driving simulator with double-wishbone suspension using efficient block-triangularized kinematic equations. *Multibody Syst. Dyn.* **28**(4), 331–347 (2012). <https://doi.org/10.1007/s11044-011-9299-y>
141. Uchida, T., McPhee, J.: Using Gröbner bases to generate efficient kinematic solutions for the dynamic simulation of multi-loop mechanisms. *Mech. Mach. Theory* **52**, 144–157 (2012). <https://doi.org/10.1016/j.mechmachtheory.2012.01.015>
142. Venrooij, J., Rath, L., Schultze, A., L'Erario, M.: Employing real-time multibody simulation in driving dynamics development. In: *DSC 2023 Europe VR: 22nd Driving Simulation & Virtual Reality Conference & Exhibition*. Antibes, France (2023)
143. Wittenburg, J., Wolz, U., Schmidt, A.: MESA VERDE — a general-purpose program package for symbolical dynamics simulations of multibody systems. In: Schiehlen, W. (ed.) *Multibody Systems Handbook*, pp. 341–360. Springer, Berlin (1990). https://doi.org/10.1007/978-3-642-50995-7_20
144. Yu, X., Mikkola, A., Pan, Y., Escalona, J.L.: The explanation of two semi-recursive multibody methods for educational purpose. *Mech. Mach. Theory* **175**, 104935 (2022). <https://doi.org/10.1016/j.mechmachtheory.2022.104935>
145. Zerbato, L., Galvagno, E., Tota, A., Mancardi, L., Velardocchia, M., Nosenzo, V., Verrilli, G., Voglino, A.: Light Commercial Vehicle ADAS-Oriented Modelling: an Optimization-Based Conversion Tool from Multibody to Real-Time Vehicle Dynamics Model. *SAE Technical Paper 2023-01-0908* (2023) <https://doi.org/10.4271/2023-01-0908>
146. Zhou, J., Liu, C., Kubenz, J., Prokop, G.: Vehicle Modeling for Dynamics Analysis Based on a New Hybrid Algorithm. *SAE Technical Paper 2017-01-1562* (2017) <https://doi.org/10.4271/2017-01-1562>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.